

Voice Until the Last Exit: Ownership Structure and the Limits of Voluntary Environmental Commitments*

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April 16, 2026

Abstract

The effectiveness of corporate climate initiatives depends not only on firms' commitments but also on existing asset ownership structure. We study this question in the context of the World Bank's Zero Routine Flaring by 2030 initiative, under which oil and gas companies pledge to eliminate routine flaring from their assets. Using asset-level ownership and satellite-based flaring measurements for nearly 20,000 oil and gas assets worldwide (2012–2024), we exploit staggered firm endorsements to study how committed firms navigate the tradeoff between voice, exercising governance influence to reduce emissions, and exit, divesting from assets that are costly to decarbonize. Committed ownership reduces flaring gradually but persistently, reaching approximately 45% below pre-commitment levels after nine years. The persistence of these gains hinges on continued committed presence: when the last committed owner divests, flaring rises by roughly 40% within two years and exceeds 90% within six years, a result we term the "last good owner effect." Retaining at least one committed voice on the board is both necessary and sufficient to sustain emissions discipline.

Keywords: Environmental regulation, corporate social responsibility, corporate governance

JEL Codes: D22, K23, K32, L71, M14, O13, Q35, Q52, Q53

*The authors gratefully acknowledge support from the Swiss National Science Foundation [grant number 192578].

1 Introduction

Oil and gas production is a major contributor to global greenhouse gas emissions, accounting for approximately 15% of total energy-related emissions worldwide (IEA [2023a](#)). Gas flaring, the burning of natural gas associated with oil extraction, is particularly significant and was responsible for nearly 1% of total energy-related greenhouse gas emissions in 2022 (IEA [2023b](#)). Beyond its climate impact, flaring represents a substantial waste of energy resources. In Sub-Saharan Africa alone, approximately 35 billion cubic meters of gas are flared annually, equivalent to half of the continent’s total power consumption. The problem is especially acute in regions with weak governance and limited gas infrastructure, where the lack of pipeline networks and processing facilities makes flaring economically rational despite its environmental and social costs.

Oil and gas companies face increasing pressure to reduce emissions from both regulatory and market sources. Recent regulatory initiatives include the U.S. EPA’s 2024 rule to curb methane emissions from oil and gas operations under the Clean Air Act and the EU Methane Regulation of 2023, which mandates leak detection and flaring limits for energy producers. Alongside these mandatory regulations, companies are increasingly adopting voluntary climate commitments. As of 2023, 80% of the largest U.S. oil and gas companies voluntarily report their Scope 1 and Scope 2 greenhouse gas emissions (Ernst & Young LLP [2024](#)). Many also subscribe to voluntary disclosure and decarbonization initiatives. A prominent example is the World Bank’s Zero Routine Flaring by 2030 initiative, launched in 2015, which has attracted endorsements from 60 oil and gas companies committing to eliminate routine flaring by the end of the decade.

Yet these commitments are made at the level of firms, while emissions occur at the level of individual oil and gas assets. Ownership of these assets frequently changes through mergers, acquisitions, and divestments, often transferring stakes between companies with very different environmental incentives and governance structures. As a result, firms that

adopt environmental commitments may reduce emissions while they operate an asset, but these improvements may not persist once ownership changes. This dynamic raises a broader question about how environmental commitments affect real emissions outcomes: do firms reduce pollution through voice, by influencing operational and investment decisions while they remain owners, or through exit, by divesting from assets that are costly to decarbonize? While divestment may improve the environmental performance of the divesting firm’s portfolio, it does not necessarily reduce emissions from the underlying assets, which may instead be transferred to owners with weaker environmental incentives. Findings from Malek et al. (2022) point toward a recent increase in the number of oil and gas deals whereby firms with net-zero commitments move assets away toward companies with no environmental commitment.

In this paper, we examine how voluntary climate commitments affect environmental outcomes when firms can both influence operations and divest from assets. We frame this question using Hirschman (1972) distinction between exit and voice as responses to organizational decline. In Hirschman’s original formulation, members of an organization who are dissatisfied with its performance can either leave (exit) or attempt to change it from within (voice), and the relative effectiveness of each strategy depends on the institutional context. We adapt this framework to the setting of jointly owned oil and gas assets, where committed equity owners face an analogous choice: they can exercise voice by using their governance rights to push for emissions reductions, or they can exit by divesting their stakes in assets that are costly to decarbonize. Because voluntary environmental initiatives rely largely on reputational incentives and stakeholder pressure, their effectiveness may depend critically on how firms navigate this tradeoff. If committed firms reduce emissions primarily through voice, improvements should persist as long as they remain involved in asset governance. By contrast, if firms rely on exit by selling high-emission assets, emissions reductions at the firm level may not translate into reductions at the asset level once ownership changes. We study these mechanisms in the context of the World Bank’s Zero Routine Flaring by 2030

initiative, under which oil and gas companies pledge to eliminate routine flaring.

The global oil and gas industry provides a particularly suitable setting to study this trade-off between voice and exit. Oil and gas fields are typically owned by multiple firms through joint venture arrangements, where one company operates the asset while other partners participate as non-operating equity owners. Although operators manage day-to-day activities, non-operating owners retain governance rights over major investment and operational decisions that can affect environmental outcomes, including flaring. At the same time, ownership stakes in oil and gas assets are frequently traded, allowing firms to divest assets that are costly to decarbonize. This combination of shared governance and active asset markets provides a natural setting to examine whether committed firms reduce emissions through influence while they remain owners or instead through divestment.

To study this, we assemble an asset-level dataset combining ownership and production data from Rystad Energy’s UCube database with satellite-based measurements of gas flaring from the Visible Infrared Imaging Radiometer Suite (VIIRS). The data track nearly 20,000 oil and gas assets across more than 120 countries between 2012 and 2024 and provide detailed information on asset ownership structures, including the identity and equity shares of each partner as well as the operator responsible for managing the field. We combine this information with satellite observations of flaring to measure emissions at the asset-year level. We focus on voluntary climate commitments under the World Bank’s Zero Routine Flaring by 2030 (ZRF) initiative, which commits companies to eliminate routine flaring. Our empirical strategy exploits two sources of variation in committed ownership: when firms endorse the initiative and when committed owners subsequently divest their stakes in assets. Using an event-study framework with asset and operator fixed effects, we examine how flaring evolves when assets become owned by committed firms and when committed ownership subsequently disappears.

We report two main findings. First, when at least one equity owner commits to the ZRF, flaring at that asset declines gradually but persistently, reaching approximately 45%

below its pre-commitment levels after nine years. This decline reflects the exercise of environmental voice by committed owners. Committed operators reduce flaring quickly by leveraging direct control over day-to-day field practices, while committed non-operators exert influence more slowly over the long run. The pattern also varies with asset characteristics: in assets already connected to gas markets, committed owners exploit existing infrastructure immediately, while in assets with no prior gas market connection, reductions are slower but ultimately more ambitious, suggesting that committed owners successfully advocate for the capital investments necessary to connect these assets to gas networks. Across both asset types and ownership roles, the evidence points to a common mechanism: committed equity owners exercise active governance voice, and that voice produces real and sustained reductions in emissions.

Second, exit by the last committed owner reverses these gains entirely. When the final committed equity partner divests, leaving no committed presence on the management board, flaring rises sharply and persistently, increasing by roughly 40% within two years and over 90% within six years relative to assets that remain under committed ownership. This is the paper’s central result, which we term the “last good owner effect.” It is not simply a divestment effect: when a committed firm exits while at least one other committed equity partner remains, flaring does not increase. The environmental discipline associated with committed ownership requires at least one committed voice on the board, not merely a history of committed ownership. Notably, this result holds even when the exiting committed firm was a non-operator, so that the same operator continues managing the field before and after the transaction. The removal of committed governance presence is sufficient to trigger deterioration, independent of any change in operational management. These findings suggest that committed firms can improve their portfolio-level emissions metrics by divesting high-emitting assets to uncommitted buyers, while actual emissions at those assets persist or worsen, a form of greenwashing through governance exit that is invisible in firm-level emissions data.

Our paper contributes to three strands of literature. First, we contribute to the literature on exit versus voice in corporate governance. Admati and Pfleiderer (2009) and Edmans (2009) formalize this tradeoff for blockholders in public equity markets, while Edmans and Manso (2011) extend these insights to settings with multiple blockholders. Broccardo, Hart, and Zingales (2022) and Gollier and Pouget (2022) apply this framework to socially responsible investors, demonstrating theoretically that voice generally dominates exit when a majority of investors have prosocial preferences. A growing empirical literature examines how institutional investors affect corporate environmental behavior, including through climate risk engagement (Krueger, Sautner, and Starks (2020)), demand for climate risk disclosure (Ilhan et al. (2023)) influence over corporate environmental policies (Kim et al. (2019)), and the relative effectiveness of engagement versus divestment in reducing carbon emissions (Kahn, Matsusaka, and Shu (2023)). This literature, however, studies voice and exit through the lens of public equity ownership, where influence operates primarily through stock prices, proxy voting, and shareholder proposals. We provide direct empirical evidence that this tradeoff has real environmental consequences in jointly owned assets, where voice operates not through stock price pressure but through board representation and capital expenditure approval rights. Critically, we show that even non-operating minority equity owners exercise meaningful governance voice over emissions, and that their exit removes a binding environmental constraint independently of any change in operational management.

Second, we contribute to the literature on whether corporate environmental commitments translate into real emissions reductions. Bolton and Kacperczyk (2025) study firm-level climate commitments and find that committing firms subsequently reduce their emissions, though the overall effect on aggregate emissions has been small. Raghunandan and Rajgopal (2024) find that signatories to the Business Roundtable’s stakeholder commitment exhibited worse environmental and labor records than peers, with no improvement after signing. Yang, Muller, and Liang (2021) show that mandatory emissions disclosure reduced US power plant emissions. In the oil and gas sector, Wen, Xiao, and Peng (2023) evalu-

ate the aggregate effectiveness of the ZRF initiative using synthetic control methods, and Chang, Christensen, and McKinley (2025) study the same initiative in African onshore fields and find that flaring declines substantially in continuously operated blocks, with larger reductions where preexisting governmental oversight is weakest. We extend this literature by showing that the persistence of commitment-driven emissions reductions depends critically on the ownership structure of the underlying assets. While commitments do reduce flaring, these gains are fully reversed when the last committed owner exits, a result that is invisible in firm-level emissions data. Our findings highlight a structural limitation of voluntary initiatives that existing studies, which typically evaluate commitments at the firm or country level, are not designed to detect.

Third, we contribute to the literature on greenwashing through asset divestitures. Duchin, Gao, and Xu (2025) and Chlond and Germeshausen (2025) show that divestitures of pollutive assets do not reduce pollution at the divested facilities, while Shive and Forster (2020) and Ben-David et al. (2021) document that asset transfers to less scrutinized or less regulated owners may worsen environmental outcomes. Our setting allows us to go further in isolating the governance mechanism. We show that the environmental consequences of divestiture depend on the ownership structure that remains after the transaction. When at least one committed equity owner stays on the board, divestiture by another committed owner has no effect on flaring. Environmental deterioration occurs only when all committed governance oversight is removed, a form of greenwashing through governance exit that is invisible in firm-level emissions data.

2 Background

2.1 Flaring and the World Bank’s Zero Routine Flaring Initiative

Oil extraction typically brings up associated natural gas as a co-product, mixed with the oil in the reservoir and separated at the surface along with other fluids. Once at the surface, producers face a choice regarding how to handle this gas. They can capture it and sell it to market through pipelines or liquefaction facilities, reinject it into the reservoir to maintain pressure and enhance oil recovery, or use it on-site to power field operations. When none of these valorisation options are economically viable or technically feasible, the remaining alternatives are to flare the gas (burn it) or vent it (release it unburned into the atmosphere). Venting is considerably more harmful than flaring because unburned methane is a far more potent greenhouse gas than the carbon dioxide produced by combustion. For this reason, routine venting is prohibited or severely restricted in most oil-producing jurisdictions (World Bank [2022](#)). Flaring is therefore the default disposal method when gas cannot be valorised.

Gas flaring thus represents both a significant environmental challenge and an enormous waste of valuable energy resources. In 2022, it accounted for nearly 1% of total global energy-related greenhouse gas emissions (IEA [2023a](#)). The problem is especially acute in developing regions: Sub-Saharan Africa alone flares approximately 35 billion cubic meters of natural gas annually, equivalent to half of the continent’s total power consumption, highlighting the opportunity cost of this practice where energy access remains limited.

Policymakers and international organizations have pursued multiple approaches to reduce gas flaring in oil production. Traditional regulatory approaches establish mandatory limits on flaring volumes or rates, enforced through penalties, fines, or production restrictions, as exemplified by the US EPA’s 2024 methane rule under the Clean Air Act and the EU’s 2023 Methane Regulation. Market-based mechanisms such as carbon pricing and emis-

sions trading schemes create economic incentives for flaring reduction, while technological solutions like improved gas capture infrastructure can make flaring economically unnecessary, though such investments require substantial capital and may be uneconomical in remote locations. Recognizing the limitations of mandatory approaches in contexts with weak regulatory capacity or where international coordination is needed, voluntary commitment frameworks have emerged as an alternative strategy, relying on reputational incentives, stakeholder pressure, and corporate social responsibility to drive behavioral change.

In response to this dual challenge of environmental harm and resource waste, the World Bank launched the Zero Routine Flaring by 2030 (ZRF) initiative at the 2015 Paris Climate Conference (COP21), adopting a voluntary commitment approach. The initiative represents a voluntary commitment framework designed to eliminate routine flaring from oil production operations worldwide by 2030. Under the terms of the commitment, endorsing companies pledge to immediately cease routine flaring at all new oil field developments and to eliminate routine flaring at existing oil fields by the 2030 deadline. Signatories are also required to report annually on their progress and flaring volumes, creating a mechanism for transparency and public accountability. Since its launch, the initiative has attracted 60 oil and gas companies as endorsers, with commitments made on a staggered basis. Table 3 lists the companies that have endorsed the ZRF policy since it was announced.

Unlike mandatory flaring regulations enforced through penalties or legal sanctions, the ZRF relies primarily on reputational incentives and stakeholder pressure to drive compliance. The World Bank publishes annual progress reports documenting each signatory’s performance, creating public visibility around companies’ environmental commitments. However, there are no binding enforcement mechanisms or financial penalties for companies that fail to meet their pledges. This institutional design reflects both the practical challenges of implementing binding international environmental agreements across diverse jurisdictions and a theory of change that emphasizes the power of public commitments, investor pressure, and corporate reputation in shaping firm behavior. Whether such soft governance mechanisms

can effectively reduce emissions in practice, particularly in contexts with weak domestic regulatory capacity, remains an empirical question.

2.2 Ownership Structure and Governance in Global Oil and Gas Industry

Oil and gas extraction typically involves significant capital requirements, geological risk, and technical expertise, leading to widespread use of joint venture structures where multiple companies share ownership of individual assets or fields. In these arrangements, equity ownership is distributed among several partners, each holding a percentage stake in the asset. One partner is designated as the operator, responsible for managing day-to-day field operations, making operational decisions, and coordinating activities on behalf of all equity owners. In many cases, the operator is also the majority stakeholder. However, operator selection may also be based on technical expertise, regulatory requirements, or contractual arrangements. The remaining partners are non-operating equity owners who share in production revenues and costs proportional to their ownership stakes but do not control operational decisions. This separation between operational control and equity ownership creates a principal-agent relationship where the operator’s decisions may not fully align with the preferences of non-operating partners (Hart [1995](#); Shleifer and Robert W. Vishny [1997b](#)).

The operator’s authority extends to critical decisions affecting environmental performance, including flaring practices. Operators determine day-to-day operational practices, establish field-level environmental management procedures, and decide when flaring is operationally necessary. However, major capital investments, such as gas capture infrastructure or flare reduction equipment, typically require approval from joint operating committees where non-operating equity owners have representation and voting rights proportional to their ownership stakes. Non-operators can leverage these governance mechanisms to influence flaring decisions, including by building consensus around reduction targets, exercising

approval rights over budgetary decisions, and advocating for emissions reduction investments (Clean Air Task Force 2025). While industry reports recognize the potential for non-operator influence, existing academic research has largely focused on operator behavior, with limited empirical evidence on whether non-operating equity owners can effectively constrain emissions through governance mechanisms.

The potential for governance conflicts becomes particularly acute when joint venture partners have heterogeneous environmental commitments or face different stakeholder pressures. A publicly-listed operator facing intense investor scrutiny over ESG performance may prioritize flaring reduction even when non-operating partners, such as National Oil Companies or private equity firms with weaker disclosure requirements, might prefer to minimize costs. Conversely, a committed non-operating equity owner may struggle to influence an uncommitted operator’s environmental practices, especially if flaring reduction requires capital investments that reduce short-term profitability. These conflicts are further complicated by the fact that infrastructure investments in gas capture benefit all equity owners collectively, creating potential free-rider problems where individual partners may prefer others to bear the costs of emissions reduction (Olson 1965; Sanford J. Grossman and Oliver D. Hart 1980b).

XXX: Reframe slightly put more focus on commitment ?

Ownership changes adds more complexity to these governance dynamics. When a committed company divests its stake in an asset, the environmental implications depend critically on who acquires that stake and whether any remaining equity owners maintain climate commitments. Strategic divestitures may allow companies to improve their portfolio-level environmental metrics while actual emissions at divested assets persist or increase under new ownership. If the exiting company was the operator, operational control transfers to a new entity that may have different environmental priorities. Even if the divesting company was a non-operating equity owner, its departure may shift the balance of influence within joint management committees. These ownership transitions create natural experiments for

examining whether voluntary climate commitments reflect genuine environmental preferences or merely respond to external pressures. If non-operating committed owners exercise meaningful governance influence over emissions, their exit from joint ventures may lead to deteriorating environmental performance at the divested assets.

3 Data

In this section, we describe our data sources and how we combine it to create a novel dataset. We also present descriptive evidence on committed and uncommitted asset ownership, flaring and other key indicators in the oil and gas industry.

3.1 Oil and gas assets

Our primary data source is the UCube database on upstream oil and gas industry obtained from Rystad Energy (hereafter Rystad). It reports annual data on over 25,242 active commercial assets (i.e., a specific area or field where companies have the rights to explore or produce) spanning over 129 countries from 2012 to 2024. The dataset contains information on annual production sold and outages by oil and gas type. We define oil as any of the following: crude oil, NGL, and condensate. Since volumes reported corresponds to production reaching the market, no recorded gas volumes does not necessarily mean no associated gas was produced. It rather means that the asset is not connected to the local gas market or does not possess gas liquefying plants that would allow selling the gas.

The dataset also reports annual cost data disaggregated into categories¹. The types of cost we use are well CAPEX and facility CAPEX. The former corresponds to "capitalized costs related to well construction, including drilling costs, rig lease, well completion, well stimulation, steel costs and materials", while the latter includes "costs to develop, install,

1. Available categories are: production OPEX, transportation OPEX, selling, general & administrative OPEX, well CAPEX, facility CAPEX, exploration CAPEX, taxes reported as OPEX, royalties, government take in nature, and income tax.

maintain and modify surface installations and infrastructure”.

From the same source, we also retrieve information on the ownership structure and operatorship of each asset, and its evolution through time. The operator of an asset is generally a single company among the owners, although in certain cases joint ventures are created between different owners to operate the asset. Only one operator is reported per year, corresponding to the company having operated the asset for the largest share of the year in the case of changing operatorship. Participation per company is reported as a percentage interest in the asset. For years during which a transaction occurs, the value reported is proportional to the share of the year during which the company held the percentage interest in the asset. This allows us to identify the exact timing of the transaction within the year.

On top of volumes, economic and ownership data, we need geographical information about assets in order to match flaring satellite observations to assets responsible for it, as described in section 3.3. We thus complement data retrieved through Rystad’s UCube with geographical data and information on wells drilled obtained in 2022 through the WellCube, from Rystad as well. This dataset contains the number of wells drilled per year per asset and the purpose of the well (production, injection, or exploration). The associated shapefile contains geographical shapes of assets when available, and a generic one when not (i.e. a circle). For assets existing in the production dataset described above absent from the shapefile, a generic shape was created based on the centroid of the asset shape (available in the first dataset). Finally, we use data made available by Rystad on wells locations to improve the precision of the matching of satellite flares observations. It contains the geographical location of more than 1,700,000 wells drilled since 2000. In some countries, the locations reported correspond to the centroid of the asset shape, while for others such as the US and Canada, each well has a unique location².

2. Onshore US (except Alaska) and Canada only containing generic shapes, this data on wells precise location is crucial.

3.2 Commitments by companies

The World Bank’s Zero Routine Flaring by 2030 provides the list of companies that endorsed the initiative. By using the Wayback Machine, we search through all snapshots between the commencement of the program and the end year of our analysis to identify when the companies committed to the initiative. We then scrape through the past versions of the website and note the year in which the company’s name appears for the first time, and we use this as the year the company endorsed the program.³ We link this to the Rystad database using the company names and use various sources to ensure that the names are consistent across both the data sources. Table 3 lists the companies and their endorsement years. Manual corrections were performed to account for joint ventures in which committed companies were involved.

3.3 Flaring

We obtain the flaring data from the Colorado School of Mines’ dataset on Global Gas Flaring Observed from Space. The data is collected by the National Oceanic and Atmospheric Administration (NOAA) using the Visible Infrared Imaging Radiometer Suite (VIIRS) instrument aboard three satellites. It can record a wide range of electromagnetic spectral bands between the wavelengths $0.412\mu m$ and $12.01\mu m$. Elvidge et al. 2015; Zhizhin et al. 2021 develop a methodology that exploits specific shortwave infrared wavelengths during nighttime satellite observations, commonly used for fire detection, to detect active flares and derive estimates of flared volumes. Using the coordinates of the flaring observations, we link this to the Rystad data using the location of the wells or the shape of the assets. The methodology in order to match satellite observations follows a five step process allowing to attribute each flaring observation to a unique asset when possible, or to split flared volumes

3. The same information can be found on the World Bank’s ZRF reporting dashboard, however, it appears that only a subset of the companies are reported here. The current list of endorsers can be found at <https://www.worldbank.org/en/programs/zero-routine-flaring-by-2030/endorsers>, while the dashboard can be accessed from <https://www.worldbank.org/en/programs/zero-routine-flaring-by-2030/reporting>.

among nearby active assets when needed. The precise methodology used to perform this matching is described in appendix. Among 141,841 observations over the 2012-2024 period, 134,580 were matched to at least one asset (representing 95.3% of total volumes flared), of which 56% were attributed to a unique asset.

3.4 Identifying divestments

From the ownership data described above, we leverage the structure of the data to identify all transactions occurring for each asset and their characteristics. The key feature of the data allowing us to retrieve this information is the proportional nature of participation interests reported. For each company within each asset, we compute the percentage interest sold or bought when possible, or derive it from other companies involved in the transaction if possible. Same goes for the timing of the transaction. Manual corrections were made in three particular cases. For assets located in 2 blocks in Nigeria (“OML 011” and “OML 017”), no transaction was recorded in the data, but a transaction was recorded in older ownership data from the same source extracted a year before. After verification, the older data was used for these assets. Verifying that the operating company was either a company holding shares of the asset or a joint venture led us to manually verify and correct the operator in specific years for 9 assets (7 in Congo, 2 in Gabon). Finally, apparent inconsistencies in the computations of implied shares and transaction timing brought manual verification and correction for 9 assets.

Overall, among the 24,159 assets active during the period, 13,155 experienced at least one transaction. For 747 assets, a large number of simultaneous and/or consecutive transactions made the computation of the share bought and sold by each company unachievable. We drop these assets from our sample. For all other assets, we verify that the computed shares sold and bought by each company, and the computed timing for each transaction, implicates a total percentage available of exactly 100%. 615 assets did not pass this verification process for at least one year, and are therefore dropped from our sample. Finally, restricting

to assets that produced and sold any oil during the period leaves us with a sample size of 19,968 assets.

For the remaining sample, commitment to the ZRF was defined as the commitment by any company owning interest in the asset, or the buying by any committed company of an interest in the asset. The commitment treatment lasts as long as any share of the asset is held by a committed company. Conversely, the divestment treatment is defined as the selling of all shares owned by one or several committed companies in an asset. This means that in the case of a progressive divestment (either a committed company progressively selling its shares, or one committed company fully divesting before the other divests), only the last committed owner exiting leads to a treatment. The divestment treatment lasts as long as no committed company holds any share of the asset. This implies that if a company owning shares in the asset commits later, after the divestment, the asset will stop being considered as treated. We also define an operator divestment treatment, which happens once an operator commits to the ZRF or when the operatorship of the asset is transferred to a committed company.

From these three main treatment variables, we define a variety of refined treatment variables allowing us to better understand the dynamics of divestment and its consequences. We first define an “incomplete” divestment variable, identifying full divestments from committed companies in assets where at least one committed owner stays. We then divide the divestment variable into two separate cases: divestment in assets in which the exiting company is also the operator and transfers operations to an uncommitted firm, and divestment in assets that were already operated by an uncommitted firm. Finally, we build a placebo treatment to test the robustness of our findings by identifying all other possible divestments: from uncommitted owners to other uncommitted owners, from uncommitted to committed ones, and finally from committed to committed firms.

Table 4 displays summary statistics for the different samples used and main treatment groups. The first half of the table reports averages for relevant variables for the complete

sample, comprising all oil producing assets, while the second half restricts the sample to assets with no recorded gas sales (indicating no connection to any gas market). For each sample, our treatment group of interest consists of assets whose owners (at least one) committed to the ZRF initiative before divesting to uncommitted owners: 672 assets fall into this category, of which 325 are not connected to gas markets before commitment. Two opposite control groups are then defined: assets that became committed and then remained committed through at least one shareholder until the end of the period (hereafter always-committed assets), and assets that never had any committed shareholder (hereafter never-committed assets).

For both production levels and flared volumes, the distribution is heavily skewed to the right in all groups, with a small number of assets being responsible for most oil production and flared gas. Unsurprisingly, assets unconnected to the gas market seem to produce less oil on average but flare larger quantities of associated gas, highlighting the key benefit of having access to an existing gas market easily.

Although no group seems to differ significantly from one another, it is worth noting that always-committed assets tend to be larger on average than other assets. Two factors can explain this discrepancy: first, larger companies are on average more often committed than smaller ones, and tend to have working interests in larger oil fields. Secondly, large assets ownership is more often split between different firms, increasing the probability that one of them is committed. Contrary to always-committed assets, the never-committed sample display average age, production levels and flared volumes remarkably close to that of divested assets, making it an ideal control group in our setting.

4 Empirical Strategy

4.1 Event Study Approach

We aim to measure how flaring volumes change when there is a shift in an asset owner’s commitment to reducing flaring. We use an event study approach to show the effects on flaring in a series of events when the share of ownership by committed firms of an asset change. We exploit two primary sources of change in the share owned by committed firms: if any of the owners endorse the Zero Routine Flaring initiative, or if any of the owners who are already endorsed sells their share of the asset to a company who is not endorsed. The former examines the effect of commitment, focusing on how the endorsed company seeks to reduce flaring—either by lowering emissions from assets it continues to own or by divesting from them. The latter captures the effect on flaring from assets that committed owners have sold, and which are subsequently owned by less committed or entirely uncommitted owners. However, to purely capture the effects of committed owners, we need to eliminate the effect of potential confounders by omitted variables. We use a combination of high-dimensional fixed effects to account for different sources of unobserved heterogeneity at the asset, year, continent, and operator levels.

We include asset fixed effects to control for time-invariant characteristics that may influence flaring levels. These characteristics include, for example, the initial oil-to-gas ratio, the type of oil extracted, and geological attributes of the reservoir such as depth and rock type. We include continent-by-year fixed effects to absorb time-varying shocks or influences that are common to all countries within a continent in a given year—for example, the effects of global oil price movements or disruptions to international supply chains that may impact continents differently. Ideally, we would prefer to use country-by-year fixed effects to account for country-specific policy shocks. However, doing so would risk absorbing the actions of national oil companies (NOCs) in countries where they own most assets, either directly or through joint ventures. Lastly, operator fixed effects control for the efficiency and

environmental outlook of the company in charge of managing the asset.

The advantages of using an event study approach are that it allows us to estimate the dynamic effects of committed owners, as well as the pre-trends, testing whether the outcomes of the assets divested under our specified treatment are likely to have evolved similarly in the absence of the treatment. For asset i at year t , we estimate using the following event study specification:

$$Y_{it} = \sum_{\substack{k=-K \\ k \neq -1}}^{K'} \beta_k D_{i,t+k} + \gamma' X_{it} + \alpha_i + \delta_t + \theta_{it} + \varepsilon_{it} \quad (1)$$

Here, α_i denotes the asset fixed effect, δ_t the year fixed effect, θ_{it} the operator fixed effect, and X_{it} a vector of controls with conformable coefficients γ' . The scalar ε_{it} represents an unobserved shock that is not correlated with the event and the error term is clustered at the asset level. Y_{it} is the outcome of interest capturing emissions, or other economic variables that may help in explaining the change in emissions. $\sum_{k=-K, k \neq -1}^{K'} \beta_k D_{i,t+k}$ captures the dynamic effects of the treatment, and allows us to see the effect on the outcome from $K \geq 0$ years before t and $K' \geq 0$ years after t . The values of K and K' are constrained by the intersection of the data period, the timing of ownership commitments, and the year in which the asset begins or ceases production; thus, making it an unbalanced panel. The estimates $\{\beta_k\}_{k=-K}^{K'}$ are graphically summarized in the event-study plots and are normalized at $k = -1$.

Our main outcome variable is the volume of flaring, measured at the asset-year level in million standard cubic feet (mmscf/y), which is the standard reporting unit. To explore potential mechanisms, we also use several additional outcomes. These include annual capital expenditures on facility infrastructure and on drilling new wells, both measured in million U.S. dollars, the total number of wells in an asset in a given year, as well as annual oil production from the asset measured in barrels per year (bbl/y). The outcome variables contain a substantial share of zero values, with the exception of oil production, since we

restrict the analysis to assets that are actively producing. Oil production is also included as a control variable in the specifications where flaring is the main outcome.

Our preferred method for estimating the event-study coefficients is Poisson pseudo-maximum likelihood (PPML). PPML accommodates zeros in the dependent variable and allows for a proportional analysis that can account for the substantial variation in differences of baseline characteristics arising from differences in oil-field size, firm size and production (Silva and Tenreyro 2011). It is also the natural choice when the dependent variable is non-negative, as it avoids the issues associated with natural logarithm and inverse hyperbolic sine transformations (Correia, Guimarães, and Zylkin 2020). This is particularly relevant when the dependent variable is flaring, investment or production, whereas for the number of wells PPML is especially appropriate because the outcome is a count variable. Another advantage of PPML is that it performs well in the presence of high-dimensional fixed effects (Correia, Guimarães, and Zylkin 2020; Bergé 2018). Under this estimator, the coefficients of interest can be interpreted as semi-elasticities, that is the percentage change in the outcome induced by a change in ownership structure.

5 Results

5.1 Commitment and Flaring

We first examine whether voluntary commitments to the World Bank’s Zero Routine Flaring initiative translate into actual flaring reductions. Figure 1 presents event study estimates comparing flaring trajectories at assets where at least one equity owner commits to ZRF (treatment group) to assets that are never owned by committed companies during our study period (hereafter “never committed” assets). The treatment year is defined as the first instance when at least owner of an asset endorses the ZRF initiative, or when a committed owner buys into the asset. The specification includes asset fixed effects to control for time-

invariant field characteristics, year-by-continent fixed effects to absorb regional shocks and trends, and operator fixed effects to account for systematic differences in operational practices across companies. We control for oil production to ensure that changes in flaring intensity reflect operational decisions rather than changes in production volumes. Standard errors are clustered at the asset level to account for serial correlation within fields over time.

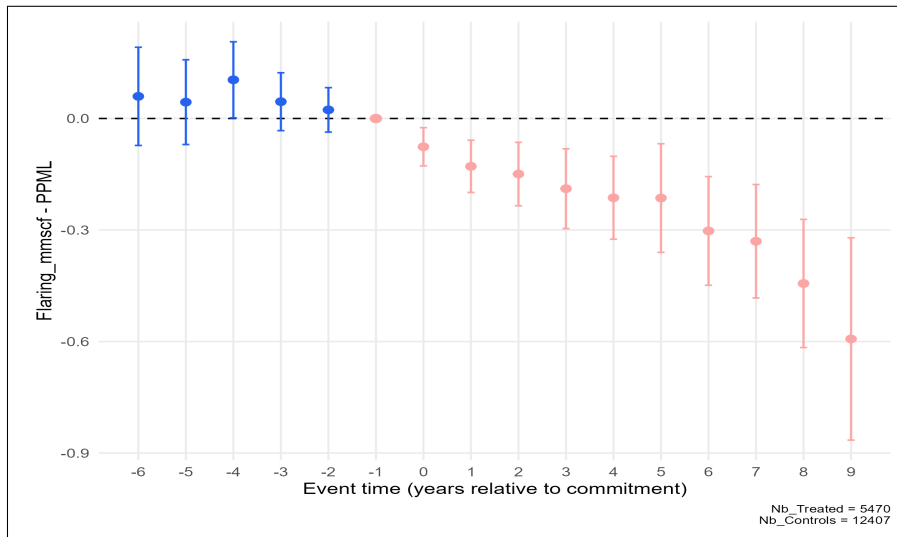


Figure 1: Event Study: World Bank ZRF commitment on flaring

Notes: This figure plots event study coefficients estimating the effect of commitment by owners of an asset on its flaring intensity. The treatment happens when at least one owner of the asset commits into the initiative, or one committed owner buys into the asset. The control group are assets which have no committed companies on their board during our study sample. Outcome is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before commitment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is $\exp(\beta) - 1$. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

The pre-period coefficients provide conclusive evidence of flat pre-trends. This flat pre-trend indicates that assets whose owners would eventually commit to ZRF were not on systematically different flaring trajectories than never-committed assets, conditional on our fixed effects and controls. Following commitment, flaring begins to decline gradually but persistently. The reductions continue to grow over time, and they are economically meaningful reductions in emissions intensity that persist through the end of our observation window. The magnitude of the last coefficient indicates around 45% of flaring is abated after 9 years⁴.

4. PPML coefficients correspond to semi-elasticities, which differ from percentage change for large coefficients. The -0.6 coefficient associated with the last point estimate is thus to be interpreted as a 45% decrease in flaring.

The persistence and magnitude of these effects indicate that voluntary climate commitments can drive substantial emissions reductions when companies maintain continuous ownership of assets.

As explained in section 4, our preferred specification includes (log) production of oil as control variable. Since our model is multiplicative, it entails that we are implicitly assessing the flaring intensity, i.e. the amount of gas flared per barrel of oil produced. However, our data only allows us to observe sold quantities of oil, which can differ from the quantities actually extracted in the asset. This explains why in numerous instances, we observe flared volumes while no oil production is recorded. These observations would be dropped if we were to use the flaring-to-oil ratio (FOR) directly as the dependent variable in our regressions. Figure 6 displays event study estimates using this specification. 330 treated assets are entirely dropped because of the exclusion of observations with no recorded oil production. Overall, estimates are particularly noisy, with most confidence intervals spanning from +2 to -2, and they do not appear to follow a specific trend.

We thus decide to keep our preferred specification, including oil production as control variable and flared volumes as dependent variable. While the observed decline in flaring observed in committed assets seems steady and sustained, it hides large disparities across treated assets that we explore below.

Pre-existing connection to gas markets. One could assume that owners of an asset already connected to gas markets could more easily abate gas flaring when required. Restricting our treated group to assets already active before becoming committed, we identify assets with prior connection to gas market as all assets in which any gas sale was recorded before commitment. Figure 2 reports event study results for both groups (connected and non-connected before commitment), as well as estimates from the whole sample. As expected, estimates confirm our intuition: the magnitude and significance of the reduction in flaring appear much larger for assets with prior connection to gas markets over the first 4 years of

commitment. Interestingly, this apparent advantage only seems play a role in the short run. Assets with no prior connection tend to catch up after five years, and to abate flaring more ambitiously afterwards. After nine years, the magnitude of the estimate is twice as large for assets with no prior connection, translating into an average decrease of 50% as compared to 30% for assets connected to gas markets before connection. Once the owner of an asset becomes committed, it appears to start using immediately existing gas selling infrastructures when there are any to reduce flared quantities, but do not seem to further invest to fully abate flaring. On the contrary, committed owners of non-connected assets require more time to invest in new infrastructures, but their actions seem on average to aim at fully abating flaring.

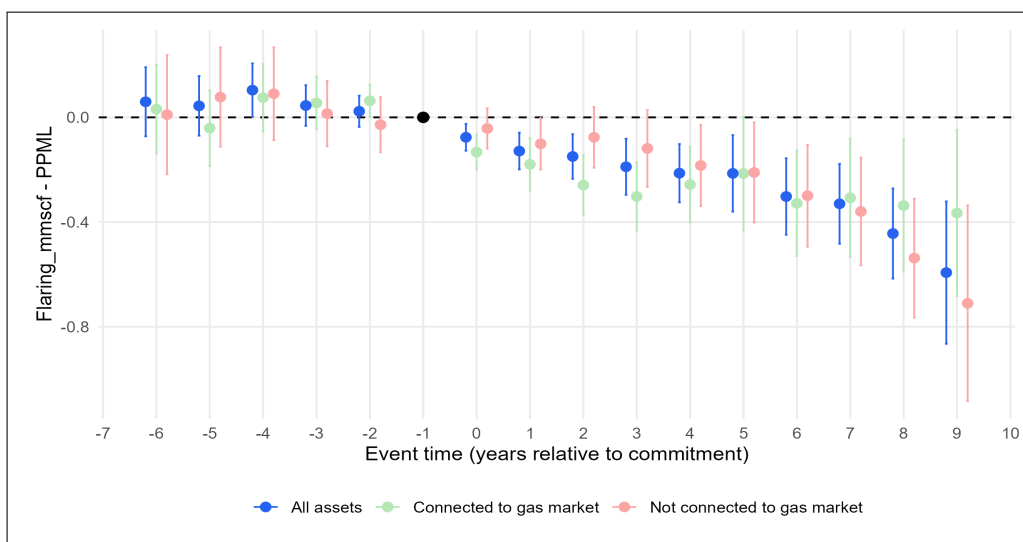


Figure 2: ZRF commitment on flaring by pre-commitment gas market connection status.

Notes: This figure plots event study coefficients estimating the effect of commitment by owners of an asset on its flaring intensity. The treatment happens when at least one owner of the asset commits into the initiative, or one committed owner buys into the asset. The control group are assets which have no committed companies on their board during our study sample. Outcome is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before commitment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is $\exp(\beta) - 1$. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

The role of operational control. Apart from assets characteristics, the dynamics among equity owners also play a role in the pace at which changes can happen within an asset. The firm operating the asset often have more authority over day-to-day practices and more

importantly over any new investments required to phase out flaring. Figure 3 displays result from the event study splitting treated assets between those in which the company committing is also the operator, and those in which the operator remains uncommitted. The difference in the short run is once again striking: committed operators tend to reduce flaring by around 15–25% immediately after commitment while committed shareholders with no operating control do not appear to affect flaring practices over the first five years of commitment. Coefficients for assets with uncommitted operator and committed shareholders eventually catch up, although late, and reflect an even larger decrease in the longer run.

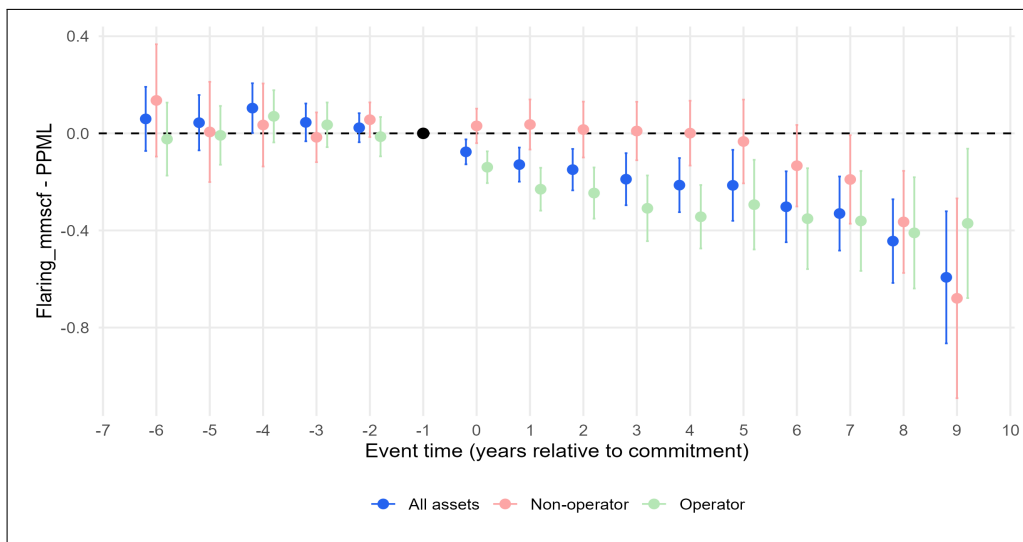


Figure 3: ZRF commitment on flaring by operator commitment status.

Notes: This figure plots event study coefficients estimating the effect of commitment by owners of an asset on its flaring intensity. The treatment happens when at least one owner of the asset commits into the initiative, or one committed owner buys into the asset. The control group are assets which have no committed companies on their board during our study sample. Outcome is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before commitment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is $\exp(\beta) - 1$. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

Commitment intensity. Decision making within the board is not only influenced by who operates the asset, but also by the relative weight of each actor. Although not operating the asset, a firm owning 50% of an asset probably has higher influence over investment decisions than an owner owning 5%. We explore this dimension by introducing in our specification an intensity of treatment, reflecting the shares owned by committed companies. Figure 4 displays the results associated with interacting treatment variables with

this intensity. The magnitude of the reduction is increased, by 10% to 50% depending on the point estimate considered. Yet, the overall evolution of flaring in committed assets is unchanged. We further look into the influence of the shares owned by committed firms by splitting our treated sample between committing firms owning the majority of the asset vs minority shares. Results in figure 7 do not allow to conclude that any significant difference exist between the two groups. Majority owners do not necessarily appear to exert more influence over the operations conducted in active assets, contrary to operating firms.

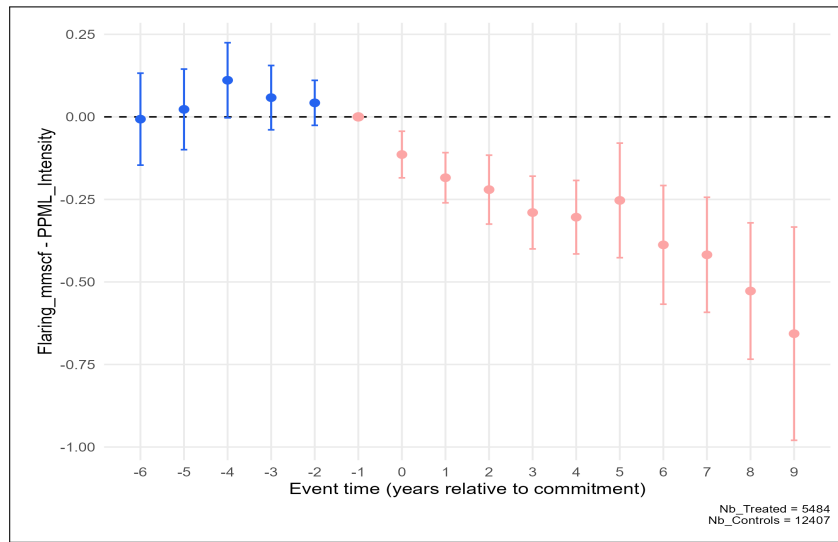


Figure 4: ZRF commitment on flaring: treatment intensity.

Notes: This figure plots event study coefficients estimating the effect of commitment by owners of an asset on its flaring intensity. The treatment happens when at least one owner of the asset commits into the initiative, or one committed owner buys into the asset. The control group are assets which have no committed companies on their board during our study sample. Dependent variable is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before commitment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is $\exp(\beta) - 1$. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

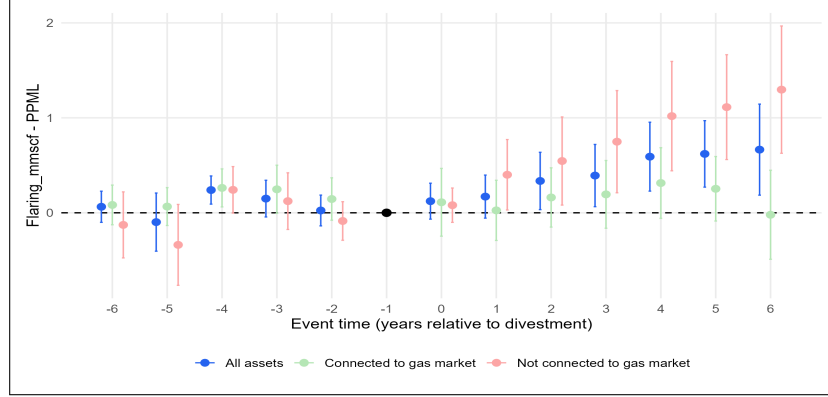
In figures 2 and 3, the specific trends and catching up by non-operators and assets with no prior connection to gas markets partly results from a selection of assets. Assets in which flaring abatement is more difficult to implement probably have a higher probability of being divested by committed owners. The results presented in this section raise important questions about what happens to flaring when committed companies divest their stakes. If the observed reductions reflect genuine changes in corporate environmental preferences and

governance priorities, we might expect flaring to increase when committed owners exit and transfer control to uncommitted parties. We examine this possibility in the following subsection 5.2. We further investigate assets characteristics as well as surrounding environment influence the decision to divest in section 5.3.

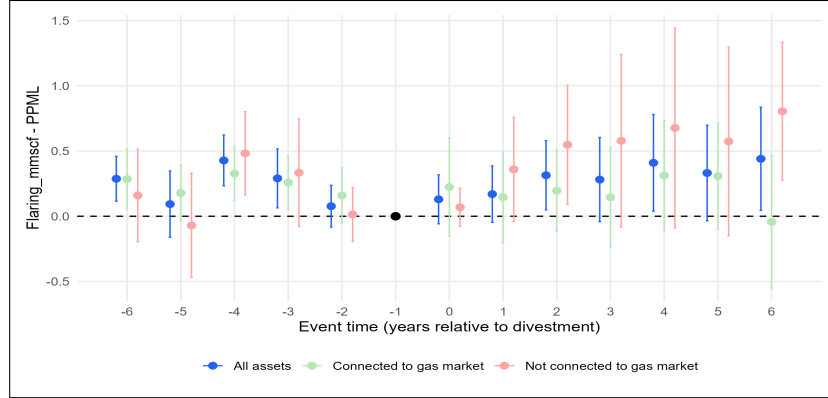
5.2 Divestment and flaring

Divestment by the last committed owner. We now examine what happens to flaring when committed companies completely divest their stakes in oil and gas assets, leaving no other committed equity owners on the management board. If committed companies were truly constraining emissions through their equity ownership and governance influence, their complete exit should lead to increased flaring under new, completely uncommitted management board. Figure 5 presents event study estimates comparing flaring trajectories at assets where the last committed owner divests (treatment group). Results are presented using two distinct control groups: assets owned by committed companies that remained in the board until the end of the period (panel A: always committed) and assets that were never owned by committed companies (panel B: never committed). The treatment year is the year when the *last* committed owner divests. The specification includes asset fixed effects, year-by-continent fixed effects, and operator fixed effects, with standard errors clustered at the asset level. Each coefficient represents the difference in flaring intensity between treated assets and control assets at a given event time, relative to one year before divestment, which is normalized to zero.

The results provide strong evidence that committed owners do keep in check fields operations regarding flaring. In the years prior to divestment, no specific trend exists between divested assets and always committed ones. When comparing to never committed assets, this pre-trend seems to go downward. This is expected, since assets with committed owners reduce flaring after signing onto the ZRF initiative. Following divestment, however,



(A) *Control group: always committed*



(B) *Control group: never committed*

Figure 5: Divestment on flaring by pre-divestment gas market connection status

Notes: This figure plots event study coefficients estimating the effect of complete divestment by committed companies on flaring intensity. The treatment happens when the last committed equity owner divests, leaving no committed companies on the management board. Dependent variable is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before divestment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is $\exp(\beta) - 1$. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

flaring increases sharply and persistently for the treated assets, as compared to both always committed and never committed assets (blue series). Since committed assets tend to see their flaring decrease progressively, it is not surprising that the magnitude of the increase observed in divested assets is sharper when comparing to always committed assets. The magnitude of this effect is economically substantial. Comparing to the always committed control group, the point estimates in years post-divestment suggest flaring increases of 40% after two years and over 90% after 6 years relative to the pre-divestment mean, indicating that the presence of committed equity owners had been meaningfully constraining emissions even after accounting for operator identity and other asset characteristics.

To put this estimate in perspective, 9% of committed assets are eventually divested, but their size is on average smaller than those staying committed, as suggested in table 4. In 2024, divested assets account for 7% of the flaring originating from oil assets committed at any point in time. This entails that the cumulative environmental gains obtained from the ZRF initiative in assets remaining committed far outweigh the adverse effects from divested assets. However, this conclusion can not be drawn locally: in divested assets, the increase in flaring post-divestment is often larger than the initial decrease post-commitment. This is due to the fact that on average, divested assets stay committed 2.7 years before being divested, limiting the magnitude of the reduction in flaring during commitment.

The role of connection to gas markets. Across both control groups, the post-divestment increase in flaring is positive but statistically insignificant for assets connected to the gas market at the time of divestment (green dots). In these assets, incoming owners have little incentive to stop gas sales, as doing so would amount to renounce to a direct revenue stream. Conversely, assets with no gas market connection at the time of divestment experience a particularly sharp increase in flaring, exceeding 200% five years after divestment when comparing to always committed assets (as suggested by the +1.1 point estimate). Moreover, evidence in the preceding subsection suggested that over the first few years of commitment, flaring is only moderately reduced in these assets. Since the average duration

of commitment in divested assets is 2.7 years, our findings suggest that for this subset of assets, the net environmental impact of the ZRF initiative is overwhelmingly negative, and can result in substantially higher quantities of gas flared as compared to a situation with no committed firm ever owning shares.

The last committed owner effect. The sharp increase in flaring following complete divestment by committed companies raises an important identification concern: are we capturing the effect of removing all committed oversight, or simply the effect of any divestment by a committed company. If our results merely reflect selection into divestment, with committed companies divesting their worst-performing or highest-emitting assets, we should observe flaring increases even when some committed companies divest while leaving other committed equity owners on the management board. Figure 8 addresses this concern by examining incomplete divestments where a committed company exits but at least one other committed equity owner remains. The specification and control groups remain identical to the complete divestment analysis. If the previous results simply captured committed companies divesting problematic assets, we should observe similar post-divestment flaring increases.

The pre-period coefficients show that treated assets exhibit lower flaring than control assets in the years before divestment, with this gap narrowing over time. The negative pre-trend coefficients, which become progressively less negative approaching the divestment event, suggest that assets experiencing incomplete divestment had been outperforming never-committed assets in emissions intensity, but this advantage was eroding. This pattern is consistent with committed companies beginning to divest from assets where their governance influence was becoming less effective or where operational conditions were deteriorating. Estimates across all post-divestment periods hover around zero or are slightly negative using both always and never committed control groups. This finding provides validation for the "last good owner effect" interpretation. In contrast to the previous result on full divestment by committed owners, this result shows that flaring either remains flat or reduces slightly

post-divestment by the committed owner. Hence, flaring increases occur specifically when all committed oversight is removed from the management board, not simply when any committed company divests. The continued presence of even a single committed equity owner appears sufficient to stop the deterioration in flaring and maintain emissions discipline through governance mechanisms. The stark contrast between complete and incomplete divestments indicates that committed non-operators exercise meaningful influence over environmental outcomes, and that this influence persists until the final committed owner exits.

The role of operational control. The main divestment results demonstrate that flaring increases when the last committed owner exits. A critical question remains: whether these effects are driven entirely by the exit of committed operators who directly control field operations, or whether committed non-operators also constrain emissions through governance mechanisms. If the baseline results simply capture the transition from committed to uncommitted operational management, we should observe effects only when committed operators divest. However, if committed non-operators exercise meaningful influence through joint operating committees, approval rights over capital expenditures, and board representation, their exits should also increase flaring even when the operator remains unchanged. To test this, we split the last committed divestment sample by whether the exiting committed company was the operator or a non-operator. The non-operator subsample provides a particularly clean test: here, the same operator remains in control before and after divestment, isolating the effect of removing committed oversight from the management board while holding operational management constant. If committed non-operators constrain emissions, we should observe flaring increases even in this subsample where operational control does not change hands.

Panel A in figure 9 presents these heterogeneous effects. All specifications use the "always committed" control group. The results reveal no significant difference in timing and magnitude. While operatorship appears to lead to a faster decrease in flaring post-commitment, it does not seem to play a role in explaining the observed increase in flaring

post-divestment. This result indicates that committed non-operators are actively constraining uncommitted operators through governance mechanisms, and their exit allows rapid operational changes even without any change in operational management. To ensure that operator fixed effects do not explain this result, Panel B reports estimates omitting these fixed effects in the model. While the magnitude of the increase in flaring decreases, excluding operator fixed effects does not change our finding: the increase in flaring post-divestment is not driven by committed operators divesting.

5.3 Potential Mechanisms

How committed owners exercise voice. We now turn to potential mechanisms that could explain why flaring decreases when an asset is at least partially owned by a committed company, and increases when it is divested. To do so, we examine how operational behaviour changes after commitment and divestment. In particular, we focus on investment and production decisions that may affect the asset’s flaring levels. Three mechanisms are investigated: investment in gas abating facilities, restriction on gas production and drilling of new wells, and connection to gas markets.

Investment in gas flaring abating technologies. We do not observe investment broken down into categories that would allow us to distinguish “green” from “brown” expenditures. However, we can analyse investment at a more aggregated level, using two broad categories: spending on facility infrastructure other than wells, and spending specifically on drilling new wells. One could expect the latter to increase after commitment, and to decrease after divestment, mirroring investments in gas abating technologies, such as gas reinjection, or gas pipelines in order to find a use for the gas. Panels A and B in figure 11 report the evolution of facility capital expenses after commitment and divestment respectively. The results indicate no meaningful change in facility-infrastructure investment following commitment or divestment, as reflected in the statistically insignificant coefficients. Overall, the point estimates are particularly noisy. This could be explained by the fact that facility

capex aggregates a wide range of capital expenses, among which only a small share would correspond to expenses aimed at phasing-out gas flaring. Moreover, many oil and gas companies do not report these expenses, translating into a modelised distribution of expenses by Rystad.

Moderating oil production. A second channel through which committed owners may reduce flaring is by moderating oil production and constraining investment in new wells. Two distinct mechanisms link production decisions to flaring intensity. In assets connected to gas markets, higher oil production translates mechanically into larger volumes of co-produced gas, which can exceed the capacity of existing midstream infrastructure. Agerton, Gilbert, and Upton [2023](#) and Allen et al. [2024](#) document this mechanism in the United States, showing that the majority of flaring in the Bakken and Permian basins arises not from a complete absence of infrastructure, but from existing gathering, processing, and transmission capacity being insufficient to handle gas volumes. In assets not connected to gas markets that cannot reinject gas, the mechanism differs. Produced gas is typically used to fuel on-site equipment, with volumes exceeding operational energy requirements flared (Schulz, McGlade, and Zeniewski [2020](#)). Since on-site energy demand is largely invariant to production levels, higher oil output increases associated gas volumes more than proportionally relative to on-site consumption, thereby raising flaring intensity. Finally, the drilling of new wells in remote locations can increase flaring intensity at the asset level: wells drilled at the frontier of existing gathering networks may not be connected to gas infrastructure at the time production begins, as operators frequently start oil extraction before completing pipeline connections (Schulz, McGlade, and Zeniewski [2020](#); Agerton, Gilbert, and Upton [2023](#)).

Figure [10](#) investigates this mechanism following commitment and divestment. For both events, the figure reports the evolution of oil production (panels A and D), well capital expenses (panels B and E), and number of wells drilled annually (panels C and F). Divestment results use never committed assets as control groups. In panels A to C, upward pre-trends

indicate that production was increasing when assets became committed through one of their owner. The production tends to stabilise after commitment, as well as investment in new wells, with well capital expenses even decreasing slightly. Despite limited magnitudes, these findings corroborate the intuition of a constraining of oil production following commitment, in order to limit excess co-produced gas.

Event studies around divestment further reinforce the importance of this mechanism. Decreasing pre-trends highlight efforts made by committed owners to limit gas flaring through this channel. Soon after divestment, wells drilling and production increase sharply, with magnitudes of the point estimates more than five times larger than the ones observed around commitment. This pattern indicates that uncommitted acquirers actively invest in production capacity following divestment, expanding drilling programs rather than focusing on flaring reduction. Combined with our main flaring results, this finding suggests that increased emissions following committed owner exits reflect deliberate reallocation of operational priorities, with new uncommitted owners massively investing in production-enhancing activities and disregarding concerns about the use or disposal of co-produced gas.

Connection to gas markets. An additional channel through which committed owners can actively reduce gas flaring is connecting the asset to gas markets. In our data, around 40% of assets active before commitment have not sold any gas at time of commitment, implying they are not connected to any gas market. In these assets, the most direct way for committed owners to voice their interests is to convince the board to invest in gas processing and gathering infrastructure in order to sell it. Initially non-connected assets that start selling gas during commitment should thus be associated with a much lower probability of being divested by committed owners. In table 1, the last column provides insights on the probability for an asset to be divested according to its gas market connection status. The regression uses cross-sectional variation across committed assets active before commitment and uses as dependent variable a dummy equal to one if the asset is eventually divested after commitment. From the coefficients, assets connected to gas markets are much less likely to

become divested, with a coefficient associated with connection during commitment twice as large as the one associated with prior connection. For reference, since 9% of committed assets are eventually divested in our sample, a 5.3 percentage point decrease in the probability of being divested implies that assets in which the connection to gas market occurs while a committed owner is in the board have a 60% lower probability of being divested than assets remaining non-connected. When gas market connections outcomes align with the voice of committed owners, the likelihood of exiting the asset is dramatically reduced.

Internal governance and divestment decisions. This last channel highlights the voice-versus-exit trade-off faced by committed owners: whether to remain on the management board and exert influence over flaring decisions, or to divest. The first two columns in table 1 provide insights on how internal governance characteristics are associated with the probability of exit by committed owners. In each regression, the dependent variable is a dummy equal to one if the committed asset is divested in year t , regressed on elements of internal governance: operator commitment status on one side, commitment intensity (i.e. the share of the asset owned by committed companies) and the number of distinct non-committed owners simultaneously on the other. Since divestment affects each of these variables, a one-year lag is introduced to reflect pre-divestment choice conditions. For reference, in each year t , approximately 1.5% of committed assets are divested.

As expected, a committed operator can exert its influence more easily than other owners, reducing the probability of exit. The coefficient associated with the variable is significant, and remarkably close to the proportion of divested assets per year, implying that exits by committed operators are scarce. Regarding the second regression, the coefficient associated with commitment intensity is significant as well, and its magnitude indicates that assets fully owned by committed owners are only rarely divested. This aligns with the idea that when committed owners own most of an asset shares, voicing their interests is easier making exit an unlikely outcome. However, the number of distinct non-committed owners is not correlated with a higher probability of divesting, as one could expect. The sign of the

coefficient is even negative, but the absence of statistical significance is consistent with no meaningful effect on the probability of exit.

	Divestment in year t		Divestment in any year
	(1)	(2)	(3)
Committed operator ($t - 1$)	-1.3e-2*** (1.5e-3)		
Commitment intensity ($t - 1$)		-1.4e-2*** (2.6e-3)	
Nb. non-committed owners ($t - 1$)		-9.8e-4 (6.8e-4)	
Connection to gas			
Before commitment			-2.7e-2*** (1.1e-2)
While committed			-5.3e-2** (2.5e-2)
Continent $\times$ Year FE	✓	✓	
Continent FE			✓
Observations	36,938	36,938	6,117
Adj. R^2 (within)	0.003	0.001	0.002

Table 1: Mechanisms: determinants of divestment probability (within assets)

Notes: Standard errors in parentheses. The sample is restricted to asset-year observations in which the asset is committed (i.e. owned by at least one ZRF signatory), including the year of divestment. $t - 1$ indicates that the variable is lagged. Column (2) includes both commitment intensity and number of non-committed owners simultaneously. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

As explained above, column (3) complements this picture by showing that connection to a gas market is associated with a significantly lower probability of divestment. Since gas market access is itself partly the product of committed owners successfully advocating for infrastructure investment, this result can be interpreted as evidence that effective voice reduces the incentive to exit. Conversely, assets where no gas connection was achieved during the commitment period may reflect governance environments in which voice proved ineffective, making divestment the more rational outcome.

External opportunities and divestment decisions. Beyond internal governance, external environment may or may not provide opportunities to facilitate the valorisation of co-produced gas, thereby influencing the decision to divest or not. Table 2 explores this dimension through three separate regressions.

In the first regression, the decision to divest in year t is regressed on the lagged share of assets in a 100km radius that have committed owners. We also include this variable

interacted with the number of assets in a 100km radius. Since gas valorisation infrastructure such as gas export pipelines typically requires the aggregation of multiple assets' production, a higher share of committed assets in an area could facilitate efforts to invest in new gas infrastructures. Results indicate that a higher share of surrounding committed assets is associated with a significantly lower probability of divestment, consistent with the idea that a denser local network of committed owners facilitates collective investment in gas valorisation infrastructure, reducing the incentive to exit. The coefficient on the interaction term is negative, as expected, but statistically insignificant, suggesting that the absolute number of surrounding assets does not meaningfully amplify this effect. This null result could reflect the relatively large geographic aggregation implied by a 100km radius, which likely captures assets that are too dispersed to plausibly coordinate on shared infrastructure investments.

We further investigate the role of external opportunities by exploiting geographical variation in gas infrastructure from the Global Gas Infrastructure Tracker ([Global Energy Monitor 2025](#)), which records the location of operating gas export pipelines, LNG terminals, and natural gas power plants worldwide, as well as projects currently under construction, under discussion, shelved, or cancelled. We aggregate these into three categories: operating infrastructure, future infrastructure (under construction or under discussion), and cancelled infrastructure (shelved or cancelled). Since the dataset provides a cross-sectional snapshot as of 2025 rather than a panel, we cannot exploit time variation in infrastructure availability. Columns (2) and (3) of table 2 therefore use a cross-sectional specification in which the dependent variable is a dummy equal to one if the asset is eventually divested by its committed owners, and zero if it remains under committed ownership throughout the sample period.

Column (2) includes dummies indicating the presence of each infrastructure category within a 50km radius of the asset, while column (3) replaces these with distance scores (computed as $100 - \text{Distance in km to the nearest infrastructure of each type}$), capturing proximity more continuously. Results are consistent across both specifications. The presence of (respectively proximity to) operating gas infrastructure is associated with a significantly lower

	Divestment in year t	Divestment in any year	
	(1)	(2)	(3)
Nearby assets (100km radius)			
Share committed ($t - 1$)	$-1.8\text{e-}2^{***}$ (3.7e-3)		
$N \times$ Share committed ($t - 1$)	$-4.9\text{e-}5$ (3.4e-5)		
Gas infrastructures dummy (50km radius)			
Operating		$-3.4\text{e-}2^{***}$ (1.1e-2)	
Future		$-3.1\text{e-}3$ (9.9e-3)	
Cancelled		$3.8\text{e-}2^{***}$ (7.7e-3)	
Gas infrastructures distance score			
Operating			$-6.8\text{e-}4^{***}$ (1.5e-4)
Future			$-7.0\text{e-}5$ (1.4e-4)
Cancelled			$6.7\text{e-}4^{***}$ (1.3e-4)
Continent $\times$ Year FE	✓		
Continent		✓	✓
Observations	37,055	7,457	7,457
Adj. R^2 (within)	0.001	0.005	0.007

Table 2: Mechanisms: determinants of divestment probability (environment)

Notes: Standard errors in parentheses. The sample is restricted to committed assets, including the year of divestment. Distance scores are strictly positive and equal to $100 - \text{Distance (in km)}$. $t - 1$ indicates that the variable is lagged. $*p < 0.1$, $**p < 0.05$, $***p < 0.01$.

probability of divestment, consistent with the idea that nearby gas infrastructure renders voice a more attractive strategy rather than exit. By contrast, proximity to cancelled infrastructure is associated with a significantly higher probability of divestment, likely reflecting environments where prior attempts to develop gas markets failed, signalling unfavourable local conditions for gas valorisation. The coefficient on future infrastructure is insignificant in both specifications, which may reflect uncertainty about whether planned projects will ultimately be completed.

Interestingly, the magnitudes of the two significant coefficients are roughly symmetric

in absolute value. A natural interpretation is that in locations where both operating and cancelled infrastructure coexist, a configuration frequently observed in capacity-constrained basins such as those in the United States, the two effects approximately cancel out, leaving the probability of divestment close to that of assets with no nearby infrastructure at all. What matters for committed owners is not merely the presence of gas infrastructure, but whether that infrastructure has sufficient capacity to absorb co-produced gas. Where existing capacity has proven insufficient, as evidenced by the existence of discussed and subsequently cancelled projects, opportunities to export and/or use co-produced gas are inexistent and exit becomes comparably likely.

6 Robustness Checks

6.1 Placebo test: divestment from uncommitted to uncommitted

A key concern with our main results is whether we are capturing the specific effects of losing committed ownership, or simply generic divestment dynamics that would occur regardless of the environmental commitments of the transacting parties. If asset divestments inherently lead to operational disruptions, management turnover, or reduced capital investment that mechanically increases flaring, our estimates would conflate these generic divestment effects with the specific role of committed ownership. To address this concern, Figure 12 examines divestments where uncommitted companies sell assets to other uncommitted companies. If our baseline results merely reflect generic divestment dynamics rather than the removal of committed oversight, we should observe similar post-divestment increases in flaring in this placebo sample.

The results provide strong validation for our main interpretation. Throughout both the pre and post-divestment periods, the estimated coefficients hover around zero with wide confidence intervals that consistently include zero. In the years following divestment, point

estimates fluctuate between small positive and negative values with no clear directional pattern. Even in the subsample of assets with no gas sales infrastructure (red series), where operational constraints make flaring reduction most challenging, we observe no systematic increase in flaring following uncommitted-to-uncommitted transactions. This contrasts sharply with our baseline results, where the exit of the last committed owner led to immediate and persistent flaring increases of 40–90%. The absence of meaningful effects in this placebo sample indicates that generic divestment dynamics do not drive substantial changes in flaring intensity. Rather, the increases we document in Section 5.2 are specifically attributable to the removal of committed governance oversight from the management board.

6.2 Alternative measures of flaring

As explained in appendix A, flared volumes used as outcome variable throughout this study is the result of a matching between satellite observations and geographical data on oil and gas assets. This matching relies on several assumptions. We test the sensitivity of our results to the two most influential ones: the distance thresholds used to allocate observations into the different stages of the matching algorithm, and the matching exponent λ . Figure 13 displays the evolution of flaring after commitment and divestment for three separate robustness checks. In panels A and B, the distance thresholds are all divided by 2. This change in assumption implies relying less on wells location data when matching satellite observations and relatively more on assets geographical shapes data. In panel C and D, a matching exponent of 1 is used instead of $4/3$. Finally, in panels E and F, a matching exponent of 2 is used.

Point estimates in the different panels suggest that our main results regarding the evolution of flaring following commitment and divestment by assets owners are remarkably consistent across assumptions on flaring matching. The post-commitment decrease in flaring is virtually identical across the different panels. Similar conclusions can be drawn for post-divestment increases in flaring, to the exception of one point estimate, corresponding to the

six years post-divestment change in flaring in panel A. The sudden drop in the coefficient, and the larger error bars can be explained by the lower number of observations participating to this point estimate. This definition of flaring can lead to allocating unique (and sometimes large volume) flaring observations to an entirely different subset of assets, and can result in introducing more noise in the data. The overall trend in post-divestment flaring practices remains unchanged excluding this single point estimate.

7 Conclusion

This paper studies how ownership structure shapes the effectiveness of voluntary environmental commitments in the global oil and gas industry. Using asset-level data on ownership and production from Rystad Energy matched with satellite-based flaring measurements from VIIRS, we track nearly 20,000 oil and gas assets across more than 120 countries between 2012 and 2024. We exploit variation in the timing of firm endorsements of the World Bank’s Zero Routine Flaring by 2030 initiative and in subsequent ownership transactions to estimate how committed ownership affects flaring at the asset level.

We report two main findings. First, assets with at least one committed equity owner experience gradual and persistent reductions in flaring, reaching approximately 45% below pre-commitment levels after nine years. Committed operators reduce flaring quickly through direct control over field practices, while committed non-operators exert influence more slowly through governance channels. Assets without prior connection to gas markets require more time to reduce flaring but ultimately achieve larger reductions, consistent with committed owners advocating for the capital investments necessary to connect these assets to gas networks.

Second, these reductions are fully reversed when the last committed owner divests. Following complete exit by committed equity partners, flaring increases sharply, rising by roughly 40% within two years and exceeding 90% within six years relative to assets that retain

committed ownership. This result, which we term the “last good owner effect,” is specific to the removal of all committed governance oversight. When a committed firm exits while at least one other committed partner remains on the board, flaring does not increase. The effect also holds when the exiting committed firm was a non-operator and the same uncommitted operator continues managing the field, indicating that committed non-operators exercise meaningful governance influence over emissions independently of operational control. A placebo test using transactions between uncommitted parties confirms that these patterns do not reflect generic divestment dynamics.

The mechanisms we document are consistent with committed owners exercising active governance voice. During their tenure, committed owners constrain production expansion and drilling activity. Uncommitted acquirers reverse these constraints rapidly after divestment, investing in production-enhancing activities while disregarding the disposal of co-produced gas. Connection to gas markets during the commitment period substantially reduces the probability of subsequent divestment, suggesting that when voice succeeds in aligning asset operations with environmental objectives, the incentive to exit diminishes. Conversely, proximity to cancelled gas infrastructure projects increases the probability of divestment, reflecting environments where prior attempts to develop gas markets failed.

These findings have implications for the design and evaluation of voluntary environmental initiatives. Standard assessments of such programs typically measure outcomes at the firm or country level, where portfolio rebalancing through divestment can improve reported emissions without reducing actual pollution. Our asset-level analysis shows that this form of greenwashing through governance exit is quantitatively important: in divested assets, the post-divestment increase in flaring often exceeds the reductions achieved during the commitment period. While the cumulative environmental gains from the ZRF initiative across assets that remain under committed ownership outweigh the adverse effects from divested assets in aggregate, this conclusion cannot be drawn locally. For the subset of assets whose committed owners eventually exit, the net environmental impact of the initiative can be

negative.

More broadly, our results suggest that ownership structure is a first-order determinant of whether voluntary commitments produce durable environmental improvements. Initiatives that rely on firm-level pledges without accounting for the governance structure of jointly owned assets risk creating incentives for committed firms to exit rather than invest, transferring pollution to less scrutinized owners. The evidence presented here indicates that retaining at least one committed voice on the management board is both necessary and sufficient to sustain emissions discipline, a finding that should inform both the design of voluntary frameworks and the regulatory oversight of asset transactions in emissions-intensive industries.

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Appendix figures and tables

Year	Companies
2015	BG, Eni, Kuwait Petroleum Corp (KPC), PetroAmazonas, Shell, SNH (Cameroon), SNPC (Congo), SOCAR, Statoil/Equinor, TotalEnergies, BP, ETAP (Tunisia), Niger Delta Petroleum Resources Ltd.
2016	Galp Energia SA, KazMunaiGaz, MOL Group, ONGC, Seven Energy Nigeria, Sonangol, Uzbekneftegaz, Wintershall / Wintershall Dea, Repsol, Nigerian National Petroleum Corporation (NNPC), Oil India Limited, Frontier Oil Limited, Pan Ocean Oil
2017	Oando Energy Resources, OMV, Petroleum Development Oman (PDO), Seplat Energy, DEA / Wintershall Dea, Lukoil, Woodside, Nile Petroleum Corporation
2018	Gazprom Neft, Sonatrach (Algeria), Petrobras
2019	KazPetrol Group, Saudi Aramco
2020	Ecopetrol, Occidental, Cairn Energy / Capricorn Energy, ConocoPhillips
2021	EOG Resources, QatarEnergy, Cepsa, Pioneer Natural Resources, Chevron, Harbour Energy, Neptune Energy, Range Resources, Petronas
2022	Hess Corporation, Savannah Energy, ExxonMobil, Vista, Civitas Resources
2024	ADNOC, Santos, Pertamina, Chord Energy, Ithaca Energy

Table 3: Year of Endorsement to ZRF by Companies

	Nb Assets	N	Mean	SD	Min	Max
World - Oil						
Committed - Divested						
Years in data	672	8,300	12.7	1.3	1	13
Current age (years)	672	8,300	34.3	17.8	1	115
Oil production (kbbl/d)	672	8,300	3.8	13.0	0	226.8
Flared volumes (mmscf/d)	672	8,300	0.6	3.5	0	117.2
Committed - Non-divested						
Years in data	6,785	69,285	11.7	2.6	1	13
Current age (years)	6,785	69,285	28.1	21.9	1	182
Oil production (kbbl/d)	6,785	69,285	8.8	59.4	0	3,435.7
Flared volumes (mmscf/d)	6,785	69,285	0.9	5.8	0	355.2
Uncommitted - Non-divested						
Years in data	12,510	131,486	11.9	2.5	1	13
Current age (years)	12,510	131,486	32.7	24.3	1	166
Oil production (kbbl/d)	12,510	131,486	3.3	15.7	0	628.8
Flared volumes (mmscf/d)	12,510	131,486	0.6	5.4	0	301.4
World - Oil - No gas sold before commitment						
Committed - Divested						
Years in data	325	3,893	12.5	1.6	1	13
Current age (years)	325	3,893	35.1	18.0	1	114
Oil production (kbbl/d)	325	3,893	3.7	14.0	0	223.8
Flared volumes (mmscf/d)	325	3,893	0.7	4.5	0	117.2
Committed - Non-divested						
Years in data	3,318	30,861	11.5	3.0	1	13
Current age (years)	3,318	30,861	31.0	23.0	1	182
Oil production (kbbl/d)	3,318	30,861	7.3	38.1	0	1,272.9
Flared volumes (mmscf/d)	3,318	30,861	1.2	8.0	0	355.2
Uncommitted - Non-divested						
Years in data	4,596	48,891	12.0	2.4	1	13
Current age (years)	4,596	48,891	33.6	21.4	1	161
Oil production (kbbl/d)	4,596	48,891	3.6	16.6	0	628.8
Flared volumes (mmscf/d)	4,596	48,891	0.8	6.0	0	269.6

Table 4: Summary statistics

Notes: This table reports summary statistics for the main samples used in the analysis. The unit of observation is the asset-year. "Nb Assets" denotes the number of distinct assets and "N" the total number of asset-year observations. "Committed – Divested" refers to assets with at least one ZRF-endorsed owner that subsequently divested to entirely uncommitted ownership. "Committed – Non-divested" refers to assets that gained at least one committed owner and retained committed ownership through the end of the sample period. "Uncommitted – Non-divested" refers to assets that never had a ZRF-endorsed owner during the sample period. The upper panel includes all oil-producing assets; the lower panel restricts the sample to assets with no recorded gas sales prior to commitment (indicating no connection to a gas market). Oil production is measured in thousands of barrels per day (kbbl/d). Flared volumes are measured in million standard cubic feet per day (mmscf/d). Current age is the number of years since the asset's first recorded production. The sample covers the period 2012–2024.

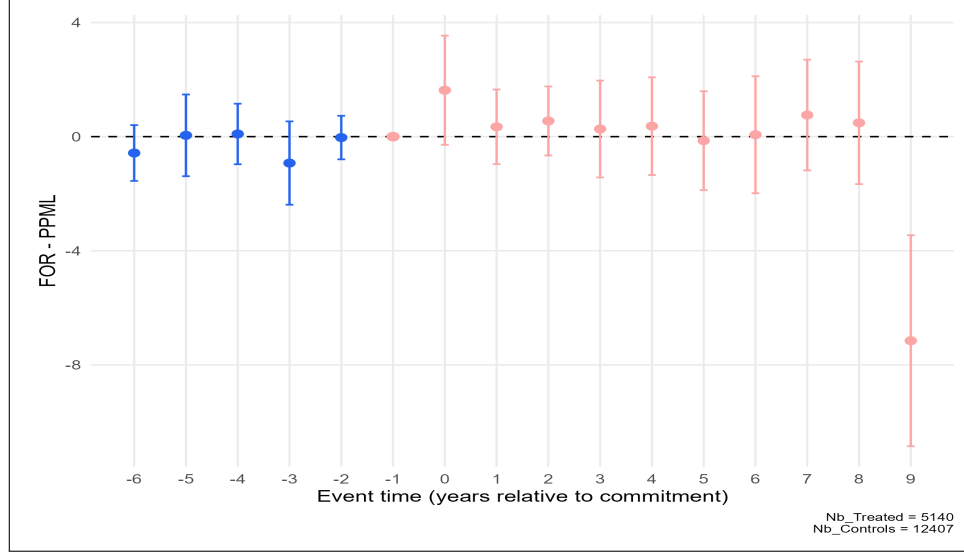


Figure 6: ZRF commitment on flaring-to-oil ratio (FOR).

Notes: This figure plots event study coefficients estimating the effect of commitment by owners of an asset on its flaring intensity. The treatment happens when at least one owner of the asset commits into the initiative, or one committed owner buys into the asset. The control group are assets which have no committed companies on their board during our study sample. Outcome is the flaring-to-oil ratio (scf of gas flared per barrel of oil). Coefficients are PPML semi-elasticities, normalized to zero in the year before commitment. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is $\exp(\beta) - 1$. Specification includes asset, year-by-continent, and operator fixed effects. Standard errors are clustered at the asset level. Point estimates are displayed if observations represent at least 10% of treated assets, and 95% error bars are reported.

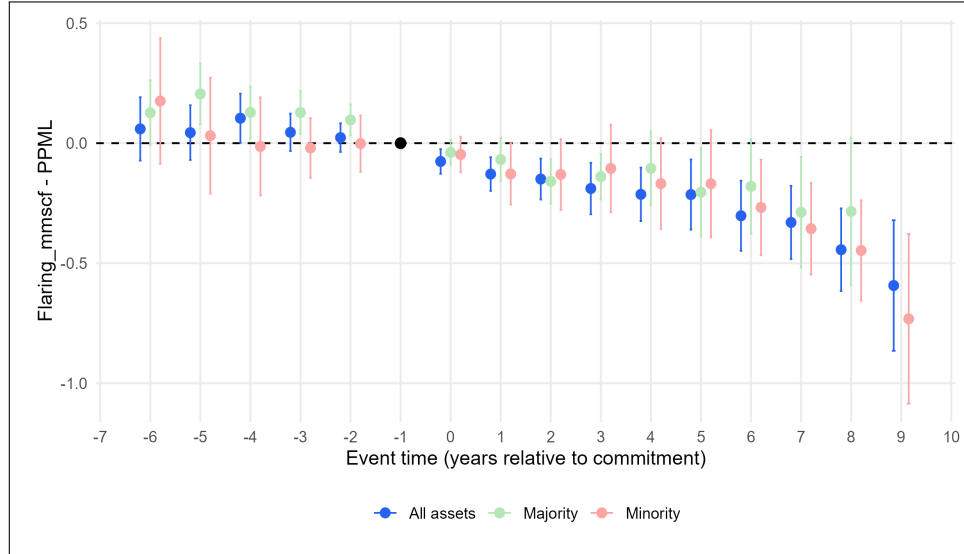
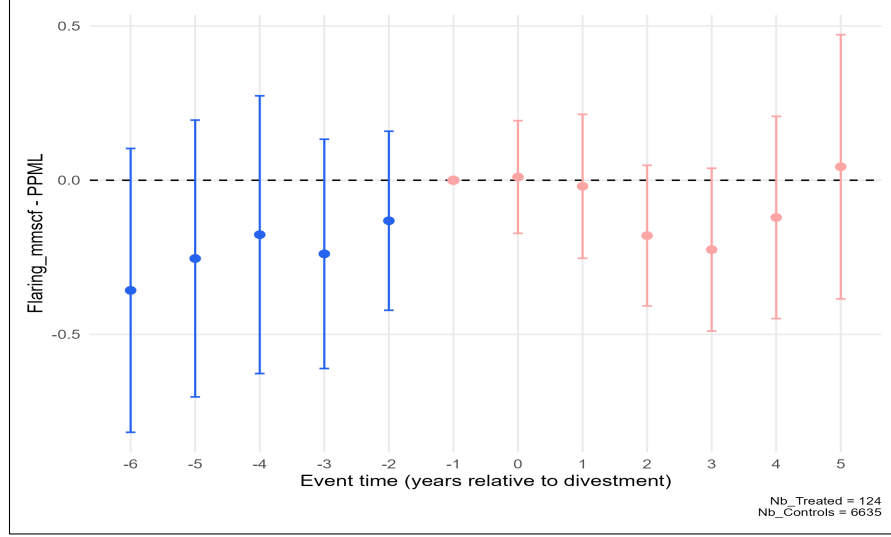
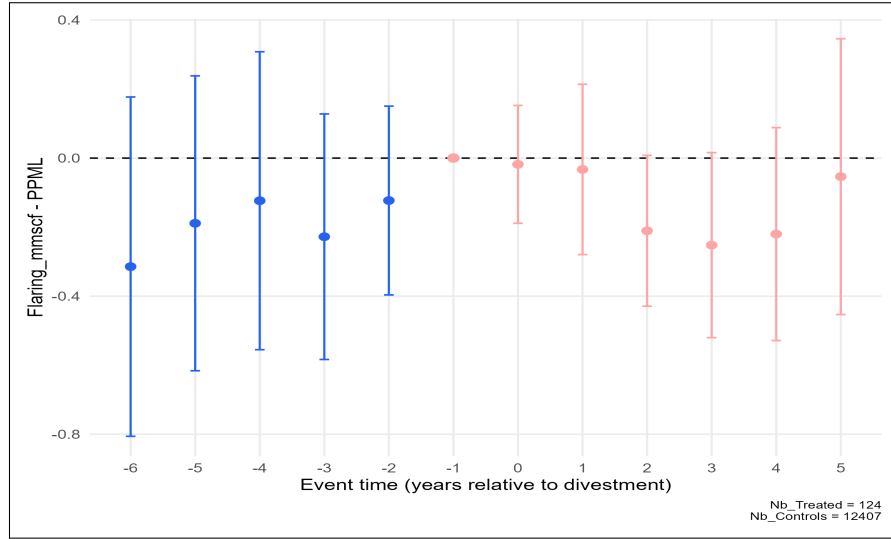


Figure 7: ZRF commitment on flaring: majority vs minority ownership.

Notes: This figure plots event study coefficients estimating the effect of commitment by owners of an asset on its flaring intensity. The treatment happens when at least one owner of the asset commits into the initiative, or one committed owner buys into the asset. The control group are assets which have no committed companies on their board during our study sample. Outcome is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before commitment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is $\exp(\beta) - 1$. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.



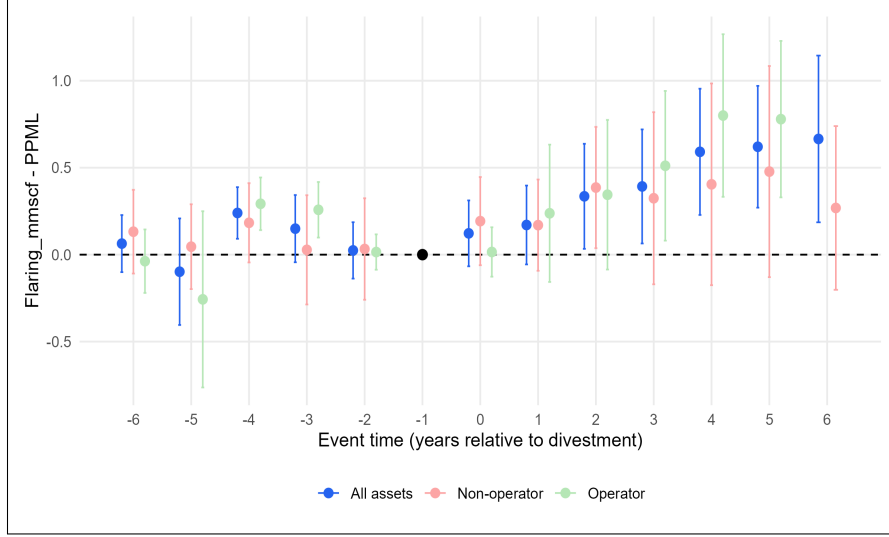
(A) Control group: always committed



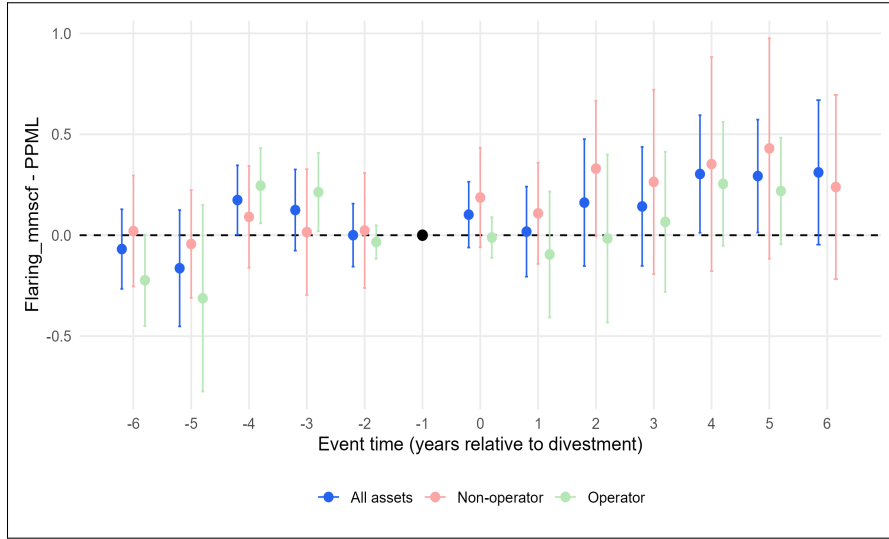
(B) Control group: never committed

Figure 8: Incomplete divestment on flaring

Notes: This figure plots event study coefficients estimating the effect of "partial" divestment on flaring intensity. The treatment happens when one committed equity owner divests, while at least one other committed firm stays in the board. Dependent variable is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before divestment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is $\exp(\beta) - 1$. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.



(A) Including operator fixed effect



(B) Excluding operator fixed effect

Figure 9: Divestment on flaring by operator commitment status.

Notes: This figure plots event study coefficients estimating the effect of complete divestment by committed companies on flaring intensity. The treatment happens when the last committed equity owner divests, leaving no committed companies on the management board. The control group are assets which remain committed until the end of the period. Dependent variable is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before divestment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is $\exp(\beta) - 1$. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

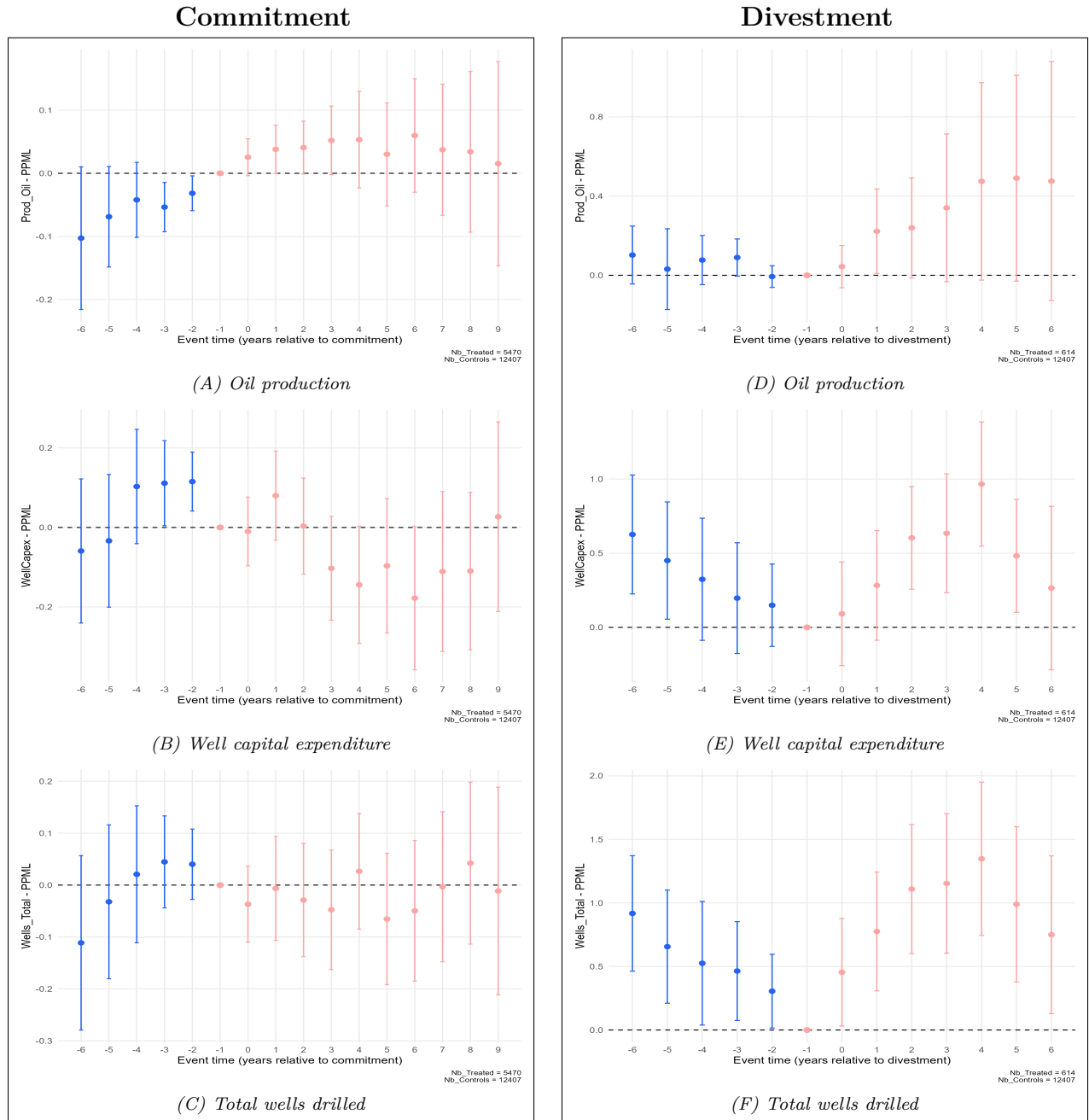


Figure 10: Mechanisms: commitment and divestment effects on production and investment

Notes: This figure plots event study coefficients estimating the effect of ZRF commitment (left) and complete divestment by the last committed owner (right) on oil production, well capital expenditure, and total wells drilled. The control group consists of assets never owned by committed companies. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceeded 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

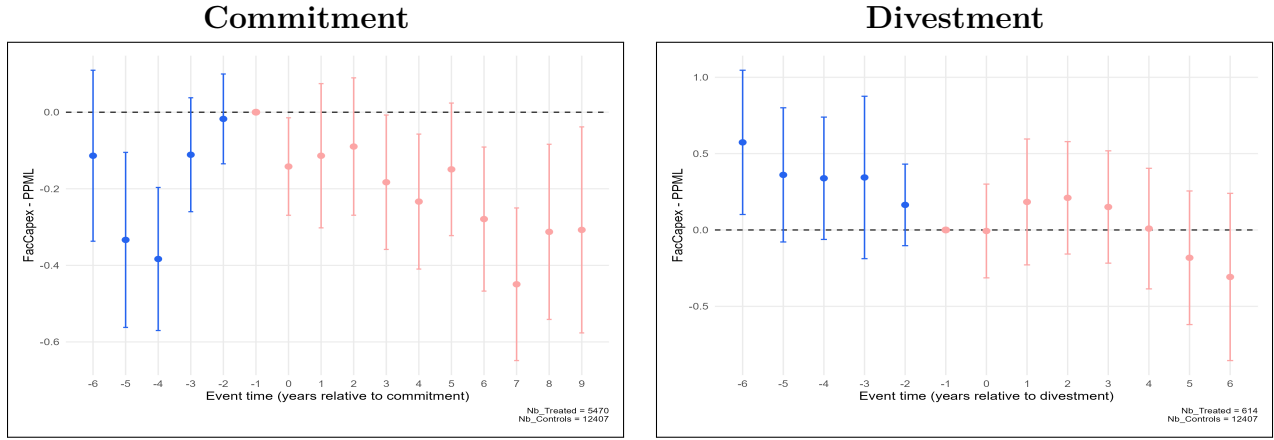


Figure 11: Commitment and divestment effects on facility capital expenditure

Notes: This figure plots event study coefficients estimating the effect of ZRF commitment (left) and complete divestment by the last committed owner (right) on facility capital expenditure. The control group consists of assets never owned by committed companies. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline; vertical bars show 95% confidence intervals.

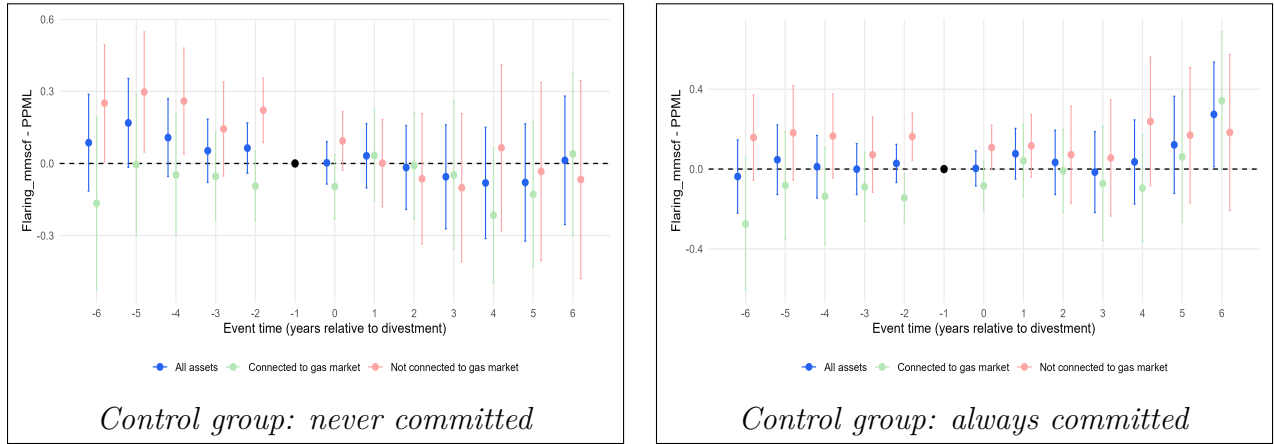


Figure 12: Divestment from uncommitted to uncommitted and flaring

Notes: These figures plots event study coefficients estimating the effect of divestment by an uncommitted owner to another uncommitted owner on flaring intensity. Dependent variable is flared volume (mmscf gas flared per year). Coefficients are PPML semi-elasticities, normalized to zero in the year before divestment. Since oil production is controlled for, they capture changes in flaring intensity. For small magnitudes, they approximate percentage changes; for larger magnitudes, the exact percentage change is $\exp(\beta) - 1$. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline. Vertical bars show 95% confidence intervals.

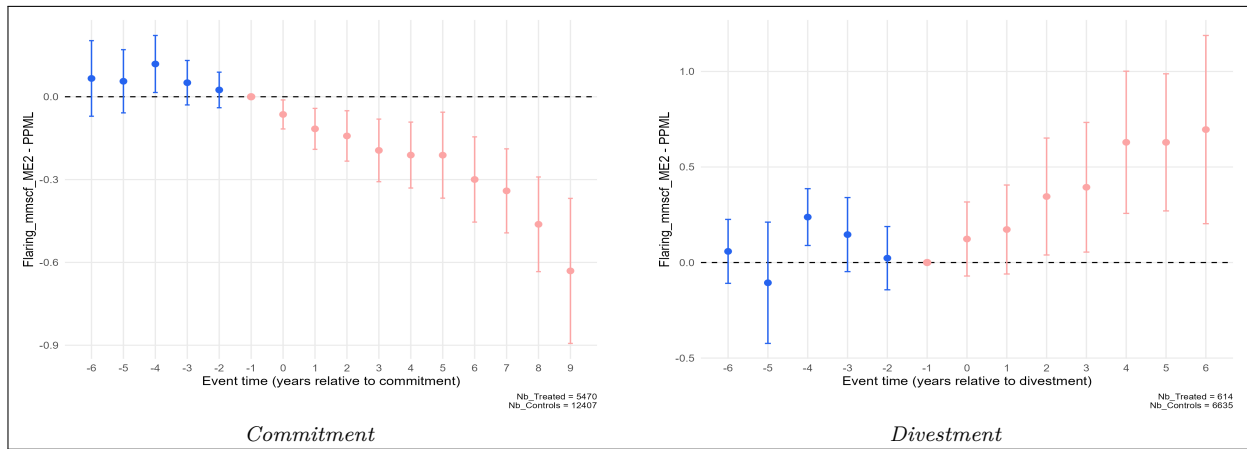
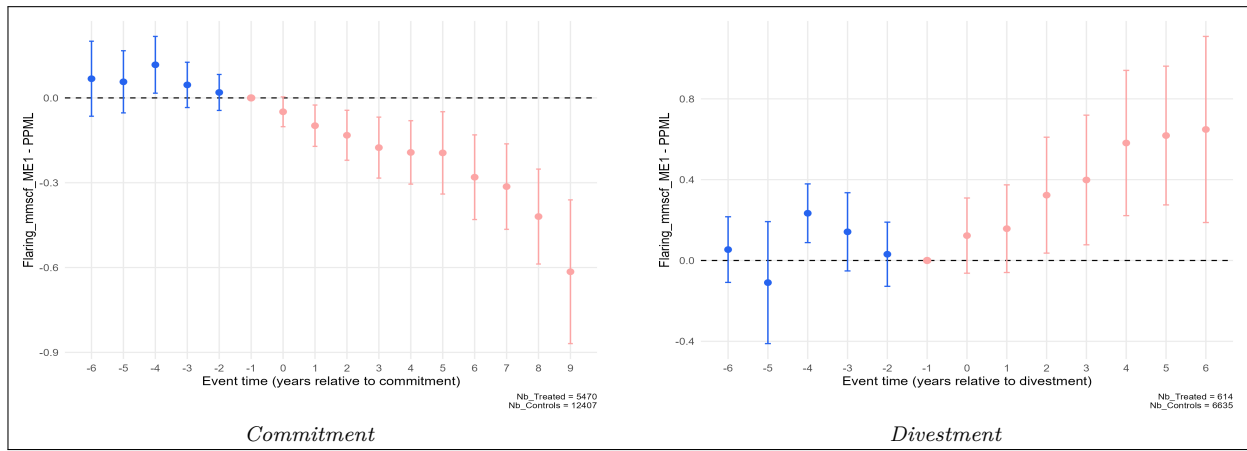
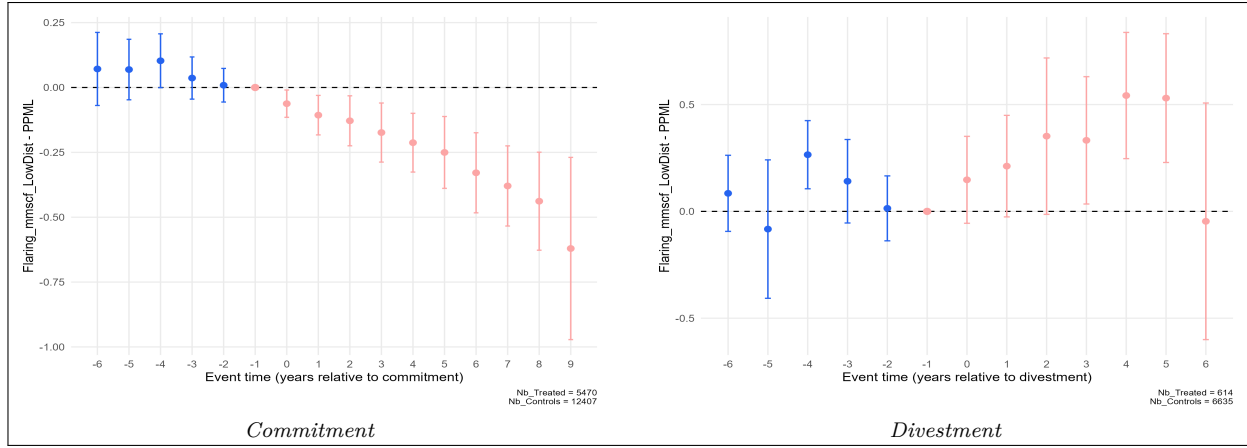


Figure 13: Commitment and divestment effects under alternative flaring measures

Notes: This figure replicates the main event study results using three alternative measures of flaring. All specifications include asset, year-by-continent, and operator fixed effects, with standard errors clustered at the asset level. Point estimates are shown only when treated observations exceed 10% of treated assets at baseline; vertical bars show 95% confidence intervals.

A Matching satellite flares observations to oil assets.

Rystad UCube database does not contain information on gas flared volumes at the asset level. This is mainly due to the fact that most oil and gas operators do not publicly report these amounts of gas flared. Moreover, under-reporting by fields operators remains an important issue making the use of self-reported flaring less reliable.

To fill this gap, we use geocoded flaring volumes calculated by the Earth Observation Group (EOG) at the Payne Institute for Public Policy, Colorado School of Mines, in partnership with NOAA. The EOG analyzes satellite data from the Visible Infrared Imaging Radiometer Suite (VIIRS), launched in 2012 and 2017, which detects heat from gas flares at night. Flaring volumes are then estimated using the VIIRS Nightfire algorithm (VNF), as detailed by Elvidge et al. 2015 and Zhizhin et al. 2021. We use the 141,841 observations with positive flared volumes between 2012 and 2024 to assign them to specific assets from Rystad’s database.

The matching relies on asset eligibility and geographic distance, using the two sources of information at our disposal described in the data section: geographic shapes of assets and geographic locations of wells drilled since 2000 within each asset.

Data on geographic shapes was available for 41,762 assets, among which 16,801 were generic. We added generic shapes for 1,916 missing assets present in our main Rystad dataset on production and ownership, based on their centroid, making the total number of generic shapes 18,785. The distribution of generic shapes is not random. For instance, onshore US (except Alaska) and Canada does not contain any defined shape. Luckily, these two countries (that represent a large share of generic shapes worldwide) are the ones in which data on well location was the most precise, making the use of this secondary database particularly useful. Moreover, some defined shapes overlap, and data on well location helped in many instances to assign flares observations to one specific asset instead of splitting flared volumes between all assets within which shapes the observation falls. The well location dataset contains 1,733,698 wells drilled since 2000, from 39,494 assets, with 857,900 unique locations (some wells being assigned the exact same location in some instances, usually the asset shape centroid).

From these two datasets, two restrictions were added. First, we tagged as ineligible assets defined as subsea tie backs. These oil and gas producing infrastructures being Under water, they cannot possess a flare themselves, and necessarily flare through another producing asset to which they are linked, usually a floating or onshore facility. This restriction removes 2,904 assets and 5,146 wells from our eligible sample. Additionally, for an asset to be eligible in a given year to be assigned a flaring observation, it needs to contain at least one well drilled and completed, whether this well is a producing well or an exploratory/wildcat one.

Once these restrictions are made, geographic distances between each flaring observations and wells or asset shapes were computed. The assignment to specific asset or the split between several follows a 3-phase procedure detailed below. For each of these procedures, a score is computed. For flaring observation o , the score of asset i is computed such as:

$$Score_{oi} = \sum (\frac{10,000}{dist_{oiw}})^{\lambda} * n_{iw}^{\kappa} + \sum (\frac{10,000}{dist_{oi}})^{\lambda}$$

Each w denotes unique wells locations of asset i , n_{iw} the number of wells in location w . κ is equal to 0.5 if w is generic (i.e. equal to asset’s shape centroid) and 1 if non-generic. Finally, λ is an exponent equal to 4/3 in the main specification⁵. We cap the lowest distances to 100m since satellite pixels definition does not allow to be more precise. The first part of the score is thus based on well locations and the second part on proximity to assets boundaries. The rationale behind the first part of the score is that the probability that an asset is responsible for a flare detected by satellite depends on both the proximity and the number of wells of an asset i near the detection location. κ is reduced to 0.5 when multiple wells locations are reported to be the asset centroid since this probably indicate that Rystad does not know the precise locations of these wells.

Attribution follows this 3-phases algorithm:

- **Phase 1:** for satellite observations with existing wells less than 2km away, only well location was used, this dataset being the most precise one. This means that only the first part of the score detailed above is used. An additional restriction is made: only wells located in a 4km radius are used to compute scores. For a given observation, once scores are computed, each asset scores are ranked from highest to lowest, and consecutive ratios are computed. If a ratio reaches 10, all following assets are eliminated. The rationale behind this selection is that if a ratio reaches such value, the probability that the observed flare actually correspond to an asset with a smaller score is extremely low. For approximately half of observations in this phase, this selection reduces the number of assets to which the observation is attributed to only one. In the remaining 50%, flared volumes are attributed to each asset by splitting proportionally to its score and its oil production capacity.
- **Phase 2:** for observations with at least one well within a 2km to 10km radius, assets shapes information is also used when possible. If at least one close-by asset has a non-generic shape, both assets shapes and wells locations are used (**phase 2A**). If all surrounding assets have generic shapes, only wells locations are used (**phase 2B**). The same process as in phase 1 is applied, except that the ratio threshold for assets with lower score to be dropped is now equal to 2. This reduction comes from the fact that scores are much lower than in phase 1 (due to the exponential nature of the

5. Robustness checks include flared volumes computed with $\lambda = 1$ and $\lambda = 2$.

score), and at these distances a ratio of 2 denotes a large difference in terms of number of wells close-by and distance to the assets' boundaries. Additionally, in phase 2A, if an observation falls within at least one asset boundaries, only these assets are kept (with a 100m tolerance), and the scores are computed among them.

- **Phase 3:** this stage applies to observation with no well in a 10km radius, but existing wells or asset boundaries in a 20km radius. If non-generic shapes exist near the observation, asset shape only matching is performed, using only the second part of the score (**phase 3A**). If only generic shapes surround the observation (**phase 3B**), the procedure is the same as the one performed in phase 2B.

Table 5 displays the number of observations matched in each phase, as well as the corresponding share of total flaring detected in terms of count and volume (2nd and 3rd columns). Phase 1 allows to match 58.3% of observations, representing 36.1% of the total volume flared. Along with phase 2A, the first phases allow to match over 70% of flaring observations worldwide. 7,261 observations remain unmatched, representing 4.7% of total flared volumes. For these observations, no completed well nor active asset boundaries was found in a 20km radius. The two last columns report, for each phase, the share of observations attributed to a unique Rystad asset. Overall, 56% of observations could be matched with a unique asset, representing 66% of total volumes.

Phase	Count	Share of total flaring		Attributed to unique asset	
		Count	Volume	Count	Volume
Phase 1	77,040	58.3%	36.1%	48.8%	64.0%
Phase 2A	24,240	18.3%	35.3%	76.6%	78.5%
Phase 2B	9,058	6.9%	5.0%	63.1%	70.9%
Phase 3A	12,088	9.1%	18.3%	49.4%	44.6%
Phase 3B	2,482	1.9%	0.6%	80.0%	80.7%
Unmatched	7,261	5.5%	4.7%	-	-

Table 5: Summary statistics: matching of satellite flares observations to oil and gas assets.

B Descriptive Example: TotalEnergies' Divestment in Gabon

To illustrate how ownership changes can affect environmental outcomes in jointly-owned oil and gas assets, we examine TotalEnergies' gradual exit from its operations in Gabon. TotalEnergies (then Total) has been a major oil and gas operator in Gabon for over 90 years, dating back to the 1950s when the company discovered major offshore deposits including the Anguille, Torpille, and Grondin fields that would become central to Gabon's oil industry. TotalEnergies, one of the largest international oil companies, is an early signatory to the World Bank's Zero Routine Flaring initiative, endorsing it in 2015. The company committed to ending routine flaring at new oil fields immediately and at existing fields by 2030, with annual public reporting on progress toward this goal.

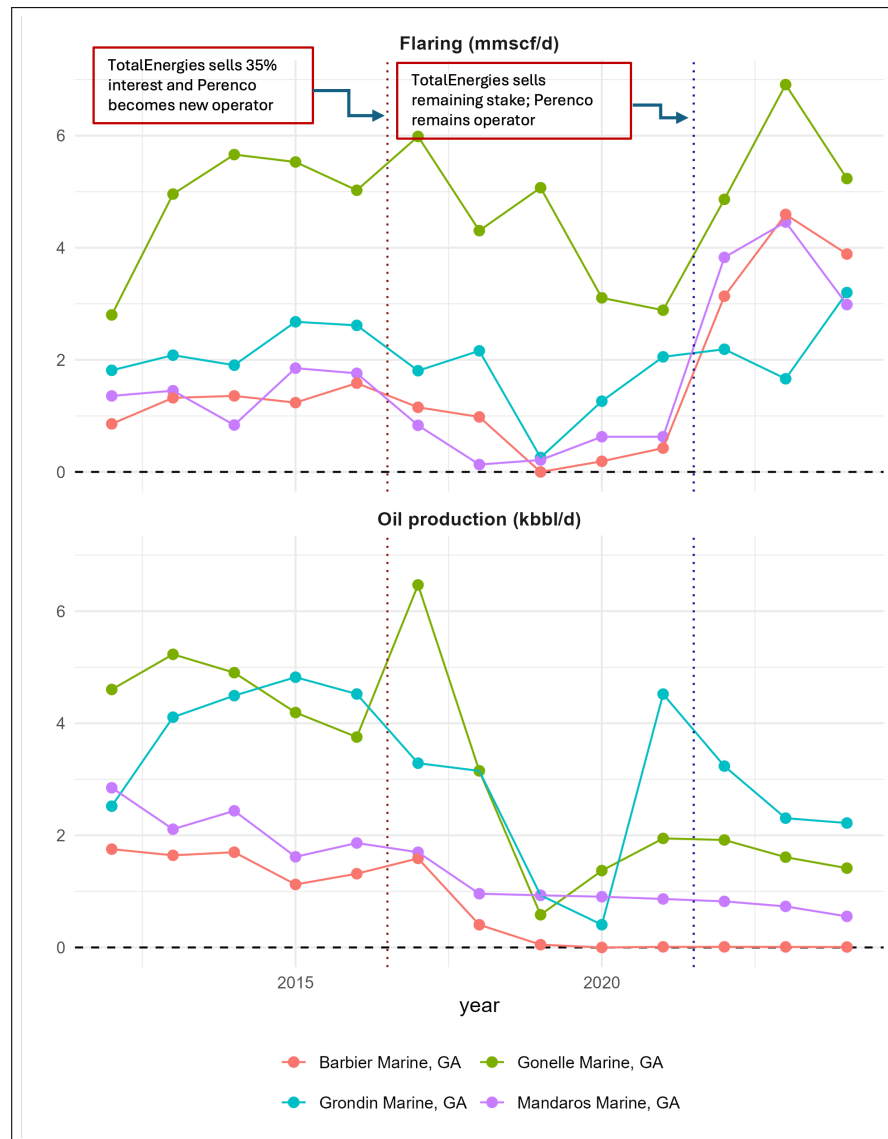


Figure 14: Flaring intensity and oil production in Gabonese marine fields before and after TotalEnergies' divestment. The first vertical line marks TotalEnergies' sale of 35% interest to Perenco and transfer of operatorship (2017). The second vertical line marks TotalEnergies' complete exit (2021).

TotalEnergies began a strategic portfolio restructuring in the late 2010s; in Gabon, they exited several offshore fields

where production had plateaued and operational costs were increasing relative to output. Total originally owned a 83% interest in these fields, the remaining 17% being owned by the Gabonese State. The divestment occurred in two stages. In 2017, TotalEnergies sold a 35% interest in four oil fields to Perenco, an Anglo-French independent oil and gas company specialized in mature fields in Africa. Crucially, this transaction also transferred operatorship of these fields from TotalEnergies to Perenco, though TotalEnergies retained a 49% stake as a non-operating partner. Four years later, in 2021, TotalEnergies, along with the State of Gabon, completed its exit by selling its remaining equity interests in these mature offshore fields to Perenco, fully relinquishing its involvement in these assets. Critically, Perenco is not a signatory to the Zero Routine Flaring initiative and, as a private company, faces substantially less disclosure pressure than publicly-listed international oil companies.

This sequence of transactions provides a particularly clean setting to examine how ownership structure affects flaring outcomes. The staggered nature of the divestment allows us to separately observe the effects of committed operator change (2017) versus complete equity exit by the committed non-operator (2021), while the transfer from a committed, publicly-listed major to an uncommitted private operator creates sharp variation in governance incentives. Figure 14 tracks annually flared volumes and oil production across these four marine fields in Gabon that experienced ownership transitions: Barbier Marine, Gonelle Marine, Grondin Marine, and Mandaros Marine.

The pattern is striking. After the first divestment when TotalEnergies gave up operatorship but held partial ownership, and Perenco was the new operator (2017-2021), flaring remained relatively stable or declined slightly. However, following TotalEnergies' complete exit in 2021, flaring increased sharply across multiple fields, even though TotalEnergies was only a non-operating owner of the asset. The effect is most dramatic in Gonelle Marine, where flaring intensity more than doubled from approximately 3 mmscf/d to over 7 mmscf/d within two years of TotalEnergies' departure. This divergence between falling or stable production and rising flaring indicates that the changes reflect operational and investment decisions rather than increased production activity, suggesting that the presence of a committed equity owner had been constraining flaring even without operational control. This increase occurred despite declining or stable oil production, suggesting the flaring results are not driven by higher production; rather this shows that *flaring intensity* changes.

This example illustrates the central questions we address in this paper. First, we examine whether committed non-operating equity owners can constrain emissions even without operational control, as suggested by the relative stability of flaring while TotalEnergies remained a minority partner. Second, we investigate whether complete exit of committed owners leads to environmental deterioration at divested assets, as evidenced by the sharp post-2021 flaring increases across all four fields. We examine these questions systematically using the universe of global oil and gas assets, exploiting variation in the timing of ZRF commitments, asset transactions, and governance structures of jointly-owned fields.

Bibliography/biblio